Medical Image Analysis (2026)

Contents lists available at ScienceDirect

Medical Image Analysis

journal homepage: www.elsevier.com/locate/media

A Hypernetwork-Driven Multi-Task Learning Framework for Diagnosis, Regression, and Prognosis in Corneal Confocal Microscopy

Hongshuo Li[a,b], Wenlong Li[c], Yalin Zheng[d,*], Hong Qi[c,*], Yitian Zhao[a,*]

[a]Ningbo Institute of Materials Technology and Engineering, Chinese Academy of Sciences, Ningbo, China
[b]University of Chinese Academy of Sciences, Beijing, China
[c]Department of Ophthalmology, Peking University Third Hospital, Beijing, China
[d]Department of Eye and Vision Science, Institute of Life Course and Medical Sciences, University of Liverpool, Liverpool, United Kingdom

ARTICLE INFO

Article history:

Corneal confocal microscopy, Neuroimmune assessment, Multi-task learning, Patient-adaptive learning, Clinical metric regression, Prognosis prediction

ABSTRACT

Corneal confocal microscopy (CCM) provides a non-invasive, cellular-resolution window into ocular and systemic neuroimmune alterations, supporting disease assessment and longitudinal monitoring. However, clinical translation remains limited by manual interpretation, invasive reference measurements, and fragmented workflows for diagnosis, continuous assessment, and prognosis. Existing CCM-based deep-learning methods mainly address isolated structures or single tasks, providing limited support for unified patient-level assessment. Moreover, unified CCM assessment faces three coupled challenges: different clinical objectives require distinct yet robust visual evidence under task conflict and cross-center variability; visually similar CCM phenotypes may correspond to different patient states; and fixed task-level predictors cannot fully accommodate patient-specific outcome mappings. To address these challenges, we propose a unified prior-guided and patient-adaptive framework for CCM-based neuroimmune assessment. It comprises: (1) relation-guided task-specific feature extraction to preserve clinically supported relations and retain task-relevant, conditionally robust visual factors; (2) patient-semantic-guided feature contextualization to incorporate patient context and resolve visually ambiguous phenotypes; and (3) patient–task-conditioned hypernetwork prediction to generate prediction parameters for each patient–task pair, enabling the final mapping to adapt jointly to patient state and task identity rather than relying on fixed task-level heads. Together, these components unify ocular neuroimmune and systemic neuropathy diagnosis, continuous clinical metric estimation, short-term treatment prognosis, and long-term postoperative prognosis. Across internal and external cohorts comprising 35,039 CCM images and 2,290 patients from eight clinical centers across China, Italy, and the United Kingdom, the proposed framework achieved state-of-the-art performance across six clinical objectives and maintained robust external performance under cross-center acquisition and cohort shifts. Beyond predictive accuracy, the estimated reference metrics preserved established clinical relationships with dry eye disease (DED)–neuropathic corneal pain differentiation, DED severity, one-month clinical outcomes, and diabetic peripheral neuropathy status. Neuro-morphometry showed progressive corneal nerve loss from healthy controls to type 1 diabetes mellitus without and with diabetic peripheral neuropathy, supporting clinically meaningful representations. These findings highlight the potential of CCM-derived digital biomarkers for integrated ocular and systemic disease phenotyping and longitudinal clinical assessment. The source code is publicly available at `https://github.com/LHS-A/CCM-DiagProg`.

1. Introduction

The cornea serves as an accessible window into neuroimmune health, with corneal confocal microscopy (CCM) enabling non-invasive, cellular-resolution visualization of the subbasal nerve plexus and resident immune cells (Gad et al., 2022; Alam et al., 2022). This microscopic phenotyping provides the biological evidence necessary to stratify disorders that share overlapping symptoms but are driven by distinct mechanisms. As visualized in Fig. 1, healthy corneas exhibit a dense, uniform nerve network. In contrast, dry eye disease (DED) is driven by inflammatory activation, marked by the prominent infiltration of immune cells (Langerhans cells). Neuropathic corneal pain (NCP) presents a structurally decoupled phenotype defined by localized neural aberrations, such as microneuromas, stemming from somatosensory dysfunction (Dieckmann et al., 2017; Moein et al., 2020). Beyond localized ocular conditions, CCM also non-invasively tracks systemic neuropathies such as diabetic peripheral neuropathy (DPN). Here, the transition from Type 1 diabetes mellitus without diabetic peripheral neuropathy (DPN−) to Type 1 diabetes mellitus with diabetic peripheral neuropathy (DPN+) is characterized by profound, progressive corneal nerve loss rather than isolated immune or neuroma events. When interpreted jointly with metabolic markers like HbA1c, these distinct structural phenotypes reliably stratify neuro-metabolic risk (Braffett et al., 2020).

To provide comprehensive patient management, however, these microscopic phenotypes must be translated beyond disease identification into quantitative clinical assessment and future outcome prediction. For ocular surface diseases, structural alterations are associated with functional endpoints such as tear-film break-up time (TBUT), corneal fluorescein staining (CFS), tear secretion (SIT), and patient-reported symptoms (OSDI) (Wolffsohn et al., 2017). Their longitudinal evolution further reflects treatment response and postoperative recovery, motivating the use of baseline CCM for both one-month treatment-response prediction and longer-term prognosis. For systemic neuropathy, joint interpretation with metabolic markers such as HbA1c provides complementary information for neuro-metabolic risk assessment (Braffett et al., 2020). Despite these biological links, current assessment remains fragmented: CCM interpretation is largely manual (Alam et al., 2022), while complementary functional and metabolic measurements require separate contact-based, dye-assisted, questionnaire-based, or blood-based procedures. A unified framework that maps CCM phenotypes to diagnosis, continuous clinical metrics, and longitudinal outcomes could therefore provide a more integrated patient-level assessment.

Recent deep-learning approaches for CCM have predominantly targeted isolated structural segmentation, e.g., corneal nerve and Langerhans cell, or end-to-end disease classification (Li et al., 2025b,c,a; Mou et al., 2022; Zhao et al., 2020; Koseoglu et al., 2025; Preston et al., 2022; Chen et al., 2024). While demonstrating technical feasibility, these methods remain largely confined to isolated anatomical structures or individual clinical objectives, limiting unified patient-level assessment. To our best knowledge, no prior automated CCM study has directly regressed clinically established continuous endpoints (CFS, TBUT, SIT, OSDI, or HbA1c). Furthermore, no framework integrates complex DED, NCP and DPN stratification, continuous metric regression, and longitudinal prognostic assessment (both short-term treatment and long-term postoperative) within a single framework. Consequently, current systems remain fragmented across isolated tasks and provide limited support for unified patient-level assessment spanning disease discrimination, quantitative clinical characterization, and longitudinal monitoring.

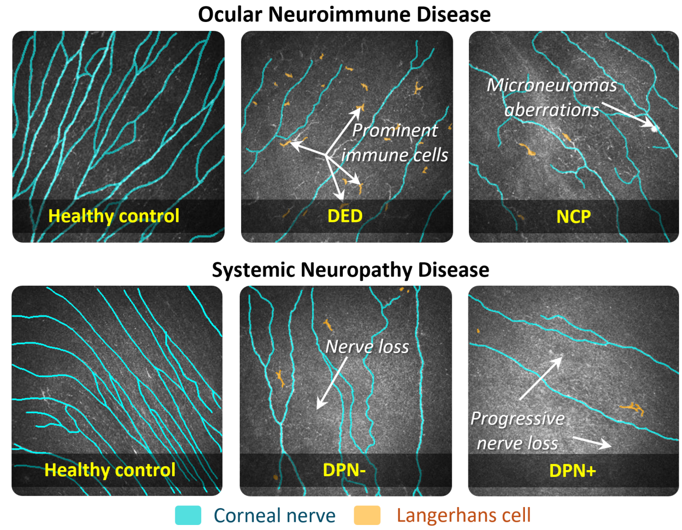

Fig. 1. Distinct microscopic imaging characteristics of ocular neuroimmune and systemic neuropathic diseases via CCM.

Developing this much-needed unified framework introduces two critical computational bottlenecks. First, *task conflicts and acquisition/cohort shifts destabilize learned features*. Categorical risk stratification, continuous regression, and prognosis necessitate heterogeneous visual evidence. Stratification requires invariant structural boundaries, while regression demands high sensitivity to localized morphological changes. Forcing these divergent objectives into a shared feature space induces gradient conflicts and negative transfer. Unmitigated acquisition variability and cross-center cohort shifts exacerbate this instability, obscuring mechanism-grounded neuroimmune markers behind imaging nuisance associations. Second, *rigid task heads fail to resolve patient-specific functional mappings*. Conventional multi-head or Mixture-of-Experts (MoE) architectures restrict sample-dependent routing to a finite set of fixed configurations, failing to dynamically adapt the prediction function to the patient's unique clinical state and objective. Overcoming these barriers requires a robust architecture capable of structurally decoupling visual evidence to resolve task conflicts, contextualizing this evidence with the patient's semantic records, and dynamically generating customized prediction parameters.

To address these clinical and computational challenges, we propose a unified, patient-adaptive framework for CCM-based ocular neuroimmune and systemic neuropathy assessment. Our

*Corresponding authors: yitian.zhao@nimte.ac.cn, Yalin.Zheng@liverpool.ac.uk, doctorqihong@hotmail.com

approach spans early warning risk stratification, continuous clinical metric regression, and longitudinal prognosis by progressively resolving heterogeneity across features, patient contexts, and prediction functions. First, relation-guided task-specific feature extraction organizes latent visual dependencies using clinically induced relations, and applies HSIC/KCI-based coarse-to-fine filtering. This allows each objective to retain its own target-relevant and conditionally robust visual factors rather than relying on a rigid shared representation. Second, patient-semantic-guided feature contextualization integrates patient-specific clinical data to reinterpret visual evidence, thereby complementing visually similar CCM phenotypes with patient-specific clinical context while preserving an image-only pathway when such information is incomplete. Third, a patient-task-conditioned hypernetwork dynamically generates custom prediction parameters for every patient-task pair. By replacing static population-level task heads with dynamically conditioned functions, the framework adapts to both heterogeneous clinical objectives and individual patient states. This enables the final mapping to adapt not only across heterogeneous objectives but also across patient states within the same objective.

The main contributions of this work are threefold:

• To our knowledge, this is the first automated CCM framework to jointly execute neuroimmune and systemic neuropathy diagnosis, alongside continuous functional metric regression, short-term treatment and long-term postoperative prognosis. By structurally decoupling multi-view features, it achieves precise stratification of healthy controls, DED, NCP, and DPN.

• The framework learns patient-adaptive mappings to directly predict established macroscopic functional outcomes (CFS, TBUT, SIT, OSDI, HbA1c) from microscopic CCM phenotypes. This reduces reliance on fragmented, contact-based measurements and enables objective neuro-metabolic risk tracking.

• We introduce a dynamic architecture that moves beyond fixed feature sharing. By combining HSIC/KCI-based feature filtering, patient-semantic contextualization, and a conditioned hypernetwork, the framework effectively resolves task interference, adapts to patient heterogeneity, and mitigates cross-center representation shifts.

2. Methods

Fig. 2 illustrates the proposed unified architecture. To support clinical CCM assessment and regression objectives, we first construct clinically induced visual relations and apply task-specific feature filtering, decoupling the required biological evidence. These visual features are then contextualized with task-aware patient semantics. Finally, a shared hypernetwork, conditioned by both patient semantics and task identity, dynamically generates prediction parameters for every patient–task pair. This mechanism enables the final prediction function to adapt seamlessly across diverse clinical objectives and unique patient states.

2.1. Relation-guided task-specific feature extraction

CCM images capture distinct neuroimmune phenotypes, but standard encoders often entangle these biological signals with imaging nuisances. Furthermore, diverse clinical objectives—spanning ocular and systemic filtering, continuous metric regression, and longitudinal prognosis—require structurally decoupled visual evidence rather than a rigid shared representation. To avoid black-box feature extraction, we organize the latent visual space using clinically induced relations grounded in experimental biological evidence. We subsequently retain multi-view factors that are highly relevant to each specific early warning task and robust against clinical confounders. This process involves three steps: visual relation representation, clinically induced relation alignment, and task-specific robust feature extraction.

2.1.1. Visual relation representation

Before introducing clinical information, we first characterize the relations already encoded by the ResNet-50 visual backbone. For the s-th CCM image I_s, the visual encoder E_{vis} extracts $F_s = E_{\text{vis}}(I_s) \in \mathbb{R}^{C\times H\times W}$, where C denotes the number of feature channels and $H \times W$ the spatial resolution. Let F_s^i denote the activation map of channel i. To reduce scale differences among channels, each activation map is normalized as $\bar{F}_s^i = (F_s^i - \min F_s^i)/(\max F_s^i - \min F_s^i + \epsilon)$, where ϵ ensures numerical stability.

While individual channels isolate specific latent responses, they do not capture the structural dependencies between distinct visual factors. To resolve this, the *feature relation modeling* block summarizes pairwise channel relations through normalized spatial co-activation across the current image mini-batch $\mathcal{B}_{\text{img}}$:

$$M_{\text{vis}}(i, j) = \frac{1}{|\mathcal{B}_{\text{img}}|} \sum_{s\in\mathcal{B}_{\text{img}}} \frac{\sum_{p=1}^{HW} \bar{F}_{s,p}^i \bar{F}_{s,p}^j}{\sqrt{\sum_{p=1}^{HW}(\bar{F}_{s,p}^i)^2}\sqrt{\sum_{p=1}^{HW}(\bar{F}_{s,p}^j)^2} + \epsilon}, \quad (1)$$

where p indexes spatial positions. The resulting $M_{\text{vis}} \in \mathbb{R}^{C\times C}$ forms the *feature relation map*, which describes the co-activation structure among latent visual factors and provides the visual-side relation representation for subsequent clinical guidance.

2.1.2. Clinically induced relation alignment

The feature relation map can contain both disease-relevant dependencies and nuisance relations induced by acquisition or cohort variation. We therefore derive a reference relational structure from available clinical supervision variables and transfer it into the latent visual space. Because clinical dependencies may be linear, monotonic, nonlinear, or information-theoretic, the *multi-view clinical relation modeling* block combines complementary dependency measures rather than relying on a single correlation criterion.

Let $C_{\text{clin}} = [c^1, \ldots, c^M] \in \mathbb{R}^{N\times M}$ denote the available clinical supervision variables in the training partition, including task-related clinical measurements and labels. Continuous variables are standardized using training-set statistics, and categorical

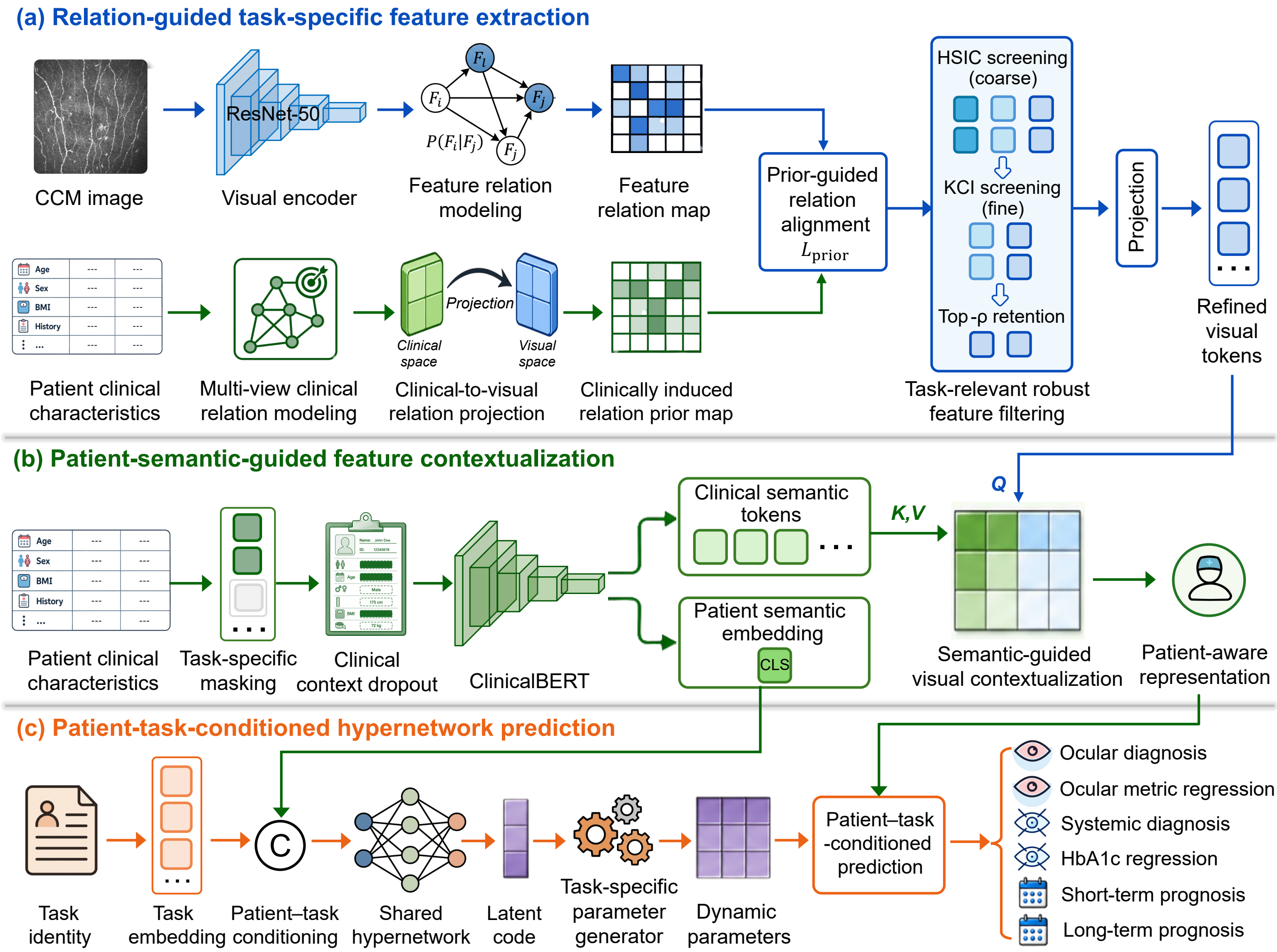

Fig. 2. Overview of the proposed framework. (a) Relation-guided task-specific feature extraction models visual feature relations, constructs a clinically induced relation prior map through multi-view clinical relation modeling and clinical-to-visual relation projection, and performs prior-guided relation alignment followed by HSIC/KCI-based task-relevant robust feature filtering. (b) Patient-semantic-guided feature contextualization applies task-specific masking and clinical context dropout before ClinicalBERT encoding, and uses the resulting clinical semantic tokens to contextualize the refined visual tokens while preserving a residual visual pathway. (c) Patient–task-conditioned hypernetwork prediction combines the patient semantic embedding with task identity and dynamically generates task-specific prediction parameters through a shared hypernetwork.

variables are one-hot encoded. Pearson and Spearman correlations are applied only to continuous, binary, or ordinal variables, while nominal categorical relations are characterized by the remaining dependency measures. NMI involving continuous variables is estimated using a k-nearest-neighbor mutual-information estimator. For each clinical-variable pair (c^m, c^r), we characterize its multi-view relation by

$$\mathbf{r}^{\text{clin}}_{m,r} = [\rho_{\text{P}}(c^m, c^r), \rho_{\text{S}}(c^m, c^r), \text{dCor}(c^m, c^r), \text{NMI}(c^m, c^r)]^T, \quad (2)$$

where ρ_{P} and ρ_{S} denote Pearson and Spearman correlations, respectively, dCor denotes distance correlation, and NMI denotes normalized mutual information. The signed correlations preserve relation direction, whereas the corresponding non-negative dependency-strength vector is $\mathbf{a}_{m,r} = [|\rho_{\text{P}}|, |\rho_{\text{S}}|, \text{dCor}, \text{NMI}]^T$, which is compatible with the non-negative visual co-activation relations used for alignment.

Different dependency measures may exhibit different stability across the training population. Each relation view is therefore estimated over 1,000 bootstrap resamples. Let v_k denote the mean bootstrap variance of relation view k across off-diagonal clinical-variable pairs. The stability weight is $\alpha_k = \exp(-v_k)/\sum_{\ell=1}^{K}\exp(-v_\ell)$, with $K = 4$, and the resulting clinical relation matrix is $A_{\text{rel}}(m, r) = \sum_{k=1}^{K}\alpha_k a^{(k)}_{m,r}$. Consequently, relation views that remain more stable under resampling contribute more strongly to the clinical relational structure.

The clinical relation matrix and the feature relation map are defined in different spaces and therefore cannot be directly compared. The *clinical-to-visual relation projection* block establishes their correspondence. After task-supervised visual warm-up, channel i is summarized as $z^i_s = \text{GAP}(F^i_s)$. For each visual feature–clinical variable pair (z^i, c^m), we compute $\mathbf{q}_{i,m} = [|\rho_{\text{P}}|, |\rho_{\text{S}}|, \text{dCor}, \text{NMI}]^T$ using the same four dependency measures. Bootstrap stability weighting gives $R(i, m) = \sum_{k=1}^{K}\beta_k q^{(k)}_{i,m}$, where β_k is estimated analogously from the stability of each feature–clinical dependency view. These association scores are normalized over the clinical dimension as

$\Pi(i, m) = \exp(R(i, m)) / \sum_{r=1}^{M} \exp(R(i, r))$.

The resulting projection transfers the clinical relational structure into the channel space and produces the *clinically induced relation prior map*:

$$M_{\text{prior}} = \Pi A_{\text{rel}} \Pi^T \in \mathbb{R}^{C \times C}. \tag{3}$$

The matrices M_{vis} and M_{prior} can now be compared in the same latent feature space. All clinical relation statistics and clinical-to-visual projections are estimated exclusively from the training partition; the resulting relation prior is fixed thereafter and requires no clinical supervision during inference.

Because diagonal elements represent self-relations rather than dependencies between distinct visual factors, they are removed before comparison. We define $O(M) = M - \text{Diag}(\text{diag}(M))$ and normalize the remaining structure as $\widehat{M} = O(M)/(\|O(M)\|_F + \epsilon)$. The *prior-guided relation alignment* block regularizes the visual relation map through

$$\mathcal{L}_{\text{prior}} = \frac{1}{C^2} \left\| \widehat{M}_{\text{vis}} - \text{sg}\left(\widehat{M}_{\text{prior}}\right) \right\|_F^2, \tag{4}$$

where $\text{sg}(\cdot)$ denotes stop-gradient. The clinically induced relation prior map therefore serves as a fixed reference structure, encouraging the visual encoder to preserve clinically supported feature relations without directly updating the reference itself.

2.1.3. Task-specific robust feature filtering

Clinically induced relation alignment regularizes the overall feature space but does not determine which channels should be retained for a particular downstream objective. This distinction is important because categorical diagnosis depends on discriminative disease boundaries, whereas continuous regression and prognosis may require sensitivity to subtler morphological variations. The *task-relevant robust feature filtering* block therefore adopts a coarse-to-fine strategy: HSIC filtering first identifies target-relevant channels, KCI filtering then evaluates whether their associations remain significant after conditioning on available non-target variables, and top-ρ retention controls the final feature budget.

For sample s, task t, and channel i, we summarize the channel response as $x_s^i = [\text{GAP}(F_s^i), \text{GMP}(F_s^i)]$, where GMP denotes global maximum pooling. Let $X^i = \{x_s^i\}_{s=1}^N$, Y_t denote the target representation of task t, and Z_t the expert-consensus-filtered admissible non-target clinical variables used for conditional filtering. For multi-output regression, all output dimensions are treated jointly as Y_t.

For the coarse filtering stage, target relevance is measured using the Hilbert–Schmidt Independence Criterion (HSIC) (Gretton et al., 2005). RBF kernels are used for continuous visual descriptors and continuous targets, whereas categorical targets use $K_{Y_t}(s, r) = \mathbb{I}(y_{s,t} = y_{r,t})$. All RBF bandwidths are estimated from the training samples using the median heuristic. With the centering matrix $H = I - \frac{1}{N}\mathbf{1}\mathbf{1}^T$ and centered kernel $\widetilde{K} = HKH$, the empirical HSIC statistic is

$$T_{\text{HSIC}}^{i,t} = \frac{1}{(N-1)^2} \operatorname{tr}\left(\widetilde{K}_i \widetilde{K}_{Y_t}\right). \tag{5}$$

Permutation-based significance is evaluated using sequential Monte Carlo permutation testing with at most 10,000 permutations and early termination once the significance decision is resolved. Feature-wise p-values are corrected using the Benjamini–Hochberg procedure at a false-discovery rate of 0.05, and the rejected channels form the target-relevant candidate set.

Target relevance alone cannot distinguish task-specific information from associations explained by other available clinical variables. The fine filtering stage therefore applies kernel conditional independence (KCI) (Zhang et al., 2011). Kernel bandwidths for Z_t are again estimated from the training samples using the median heuristic. Let $\widetilde{K}_{Z_t}$ denote its centered kernel. Generalized cross-validation determines the residualization coefficient $\widehat{\lambda}_{\text{KCI}}$, yielding $R_{Z_t} = I - \widetilde{K}_{Z_t}(\widetilde{K}_{Z_t} + \widehat{\lambda}_{\text{KCI}} I)^{-1}$.

For channel i, the joint feature–conditioning kernel is $K_{\ddot{X}^{i,t}} = K_i \odot K_{Z_t}$, where $\odot$ denotes the Hadamard product. After residualization, $K_{\ddot{X}^{i,t}|Z_t} = R_{Z_t} \widetilde{K}_{\ddot{X}^{i,t}} R_{Z_t}$ and $K_{Y_t|Z_t} = R_{Z_t} \widetilde{K}_{Y_t} R_{Z_t}$. The corresponding conditional dependence statistic is

$$T_{\text{KCI}}^{i,t} = \frac{1}{N} \operatorname{tr}\left(K_{\ddot{X}^{i,t}|Z_t} K_{Y_t|Z_t}\right). \tag{6}$$

KCI significance is evaluated using the same sequential permutation setting and Benjamini–Hochberg false-discovery-rate control. Let $\mathcal{S}_t^{\text{HSIC}}$ and $\mathcal{S}_t^{\text{KCI}}$ denote the corresponding rejection sets. Channels supported by both criteria constitute $\mathcal{S}_t^{\text{sig}} = \mathcal{S}_t^{\text{HSIC}} \cap \mathcal{S}_t^{\text{KCI}}$.

The final *top-ρ retention* step retains at most a fraction ρ of the original channels. If $|\mathcal{S}_t^{\text{sig}}| \leq \rho C$, all jointly significant channels are retained; otherwise, they are ranked according to $T_{\text{KCI}}^{i,t}$ and the highest-ranked ρC channels are selected. The resulting task-specific subset $\mathcal{S}_t$ produces $F_{\text{ref},s,t} = F_s[\mathcal{S}_t, :, :] \in \mathbb{R}^{C_{r,t} \times H \times W}$, where $C_{r,t} = |\mathcal{S}_t|$ may vary across tasks. The retained features are subsequently flattened and mapped by the task-specific projection $W_{\text{in},t}$ into the shared d_a-dimensional token space, ensuring a fixed representation dimensionality for subsequent contextualization. Thus, the common visual encoder retains task-dependent feature subsets while producing dimensionally consistent refined visual tokens.

2.2. Patient-semantic-guided feature contextualization

Task-specific filtering determines which visual factors should be retained, but the same CCM appearance can still correspond to different patient states. Patient age, systemic status, ocular history, and symptom context can therefore provide complementary information for interpreting the selected visual evidence. We introduce a patient-semantic branch that first constructs task-appropriate patient representations from the available clinical characteristics and then uses these semantics to contextualize the refined visual tokens. The residual design preserves an active visual pathway when patient information is incomplete or unavailable.

2.2.1. Task-aware patient semantic encoding

Unlike the training-only relation prior, the patient-semantic branch remains active during inference and therefore uses only information available at the intended prediction time. To prevent target leakage and maintain consistent training–inference conditions, we construct a task-safe patient context through *task-specific masking*. For patient s and task t, the *patient clinical characteristics* available at the intended prediction time are processed according to this masking strategy. Eligible information may include demographic characteristics, disease history, systemic conditions, ocular characteristics, and free-text symptoms. Three complementary constraints determine which fields can be used for each task: the direct prediction target is excluded, variables unavailable at the intended prediction time are removed, and variables directly involved in reference-label construction or otherwise revealing the requested outcome are withheld. The admissible task-specific context can be written compactly as $C_{s,t}^{\text{safe}} = C_s^{\text{avail}} \setminus (C_t^{\text{target}} \cup C_t^{\text{future}} \cup C_t^{\text{proxy}})$. The task-specific field sets are fixed before model training. Potential proxy variables are independently reviewed by three senior clinicians, each with more than 15 years of clinical experience, and are excluded when unanimous consensus determines that they could directly reveal or strongly encode the corresponding prediction target.

Clinical records may nevertheless contain different degrees of missing information. The *clinical context dropout* block explicitly exposes the semantic branch to this variability. For a patient–task pair with $K_{s,t}$ admissible clinical fields, the number of removed fields is sampled as $k_{s,t} \sim \text{Uniform}\{0, \dots, K_{s,t}\}$, after which $k_{s,t}$ fields are selected uniformly without replacement. The surviving fields are serialized into a standardized clinical sentence $S_{s,t}$ using a fixed linguistic template, while numerical values are retained rather than discretized.

The resulting sentence is encoded by *ClinicalBERT* (Alsentzer et al., 2019), $H_{\text{txt},s,t} = E_{\text{txt}}(S_{s,t}) \in \mathbb{R}^{L\times d_t}$, where L denotes token length and d_t the text embedding dimension. The token-level output $H_{\text{txt},s,t}$ forms the *clinical semantic tokens*, whereas the corresponding [CLS] representation $h_{s,t}^{\text{cls}} \in \mathbb{R}^{d_t}$ forms the *patient semantic embedding*. The former provides token-level context for visual interpretation, whereas the latter provides a compact patient-level condition for subsequent prediction adaptation.

When all admissible clinical fields are removed, $H_{\text{txt},s,t} = \mathbf{0}$ and $h_{s,t}^{\text{cls}} = \mathbf{0}$. Consequently, the same semantic pathway supports complete, partial, and absent patient clinical information.

2.2.2. Patient-semantic-guided visual contextualization

The task-specific feature extractor identifies which visual factors are relevant, whereas patient semantics provide the context in which those factors should be interpreted. The *semantic-guided visual contextualization* block therefore uses the refined visual tokens as queries and the clinical semantic tokens as keys and values.

The selected feature tensor is flattened into spatial tokens as $X_{\text{vis},s,t} = \text{Flatten}_{\text{spatial}}(F_{\text{ref},s,t})^T$ and projected into a common attention space by $X_{s,t} = X_{\text{vis},s,t} W_{\text{in},t} \in \mathbb{R}^{HW\times d_a}$. Patient-semantic contextualization is then performed through the residual cross-modal operation

$$X_{s,t}^{\text{ctx}} = \text{LN}\left[X_{s,t} + \text{Softmax}\left(\frac{(X_{s,t}W_Q)(H_{\text{txt},s,t}W_K)^T}{\sqrt{d_a}}\right)(H_{\text{txt},s,t}W_V)\right], \quad (7)$$

where W_Q, W_K, and W_V are learnable projections and $\text{LN}(\cdot)$ denotes layer normalization. Average pooling over the contextualized tokens yields $f_{s,t} = \text{AvgPool}(X_{s,t}^{\text{ctx}}) \in \mathbb{R}^{d_a}$, which constitutes the *patient-aware representation*.

The residual pathway is retained throughout contextualization. When clinical information is unavailable, $H_{\text{txt},s,t} = \mathbf{0}$, the semantic contribution becomes zero while the visual pathway remains active. This residual contextualization allows patient semantics to complement, rather than replace, the task-specific visual evidence. The resulting representation therefore encodes both task-relevant visual evidence and the available patient context, while the final mapping from this representation to the clinical outcome is handled separately by the patient–task-conditioned prediction stage.

2.3. Patient–task-conditioned hypernetwork prediction

The patient-aware representation provides individualized visual evidence, but a conventional task-specific head would still apply the same population-level mapping to all patients assigned to that objective. This remains restrictive when different patient states require different mappings from similar representations to the final outcome. We therefore condition the prediction function jointly on *patient semantics* and *task identity*, allowing a shared hypernetwork to dynamically generate prediction parameters for each patient–task pair rather than relying on fixed task-level predictors.

The framework considers six task identities, $\mathcal{T} = \{\text{oc-diag}, \text{sys-diag}, \text{oc-reg}, \text{hba1c}, \text{short}, \text{long}\}$, corresponding to ocular neuroimmune diagnosis, systemic neuropathy diagnosis, ocular-surface reference metric regression, HbA1c regression, short-term treatment prognosis, and long-term postoperative prognosis. The *task identity* is represented by a learnable *task embedding* $e_t \in \mathbb{R}^{d_e}$.

The *patient–task conditioning* block concatenates the task embedding with the patient semantic embedding as $p_{s,t} = [e_t \,\|\, h_{s,t}^{\text{cls}}]$. A *shared hypernetwork* $\mathcal{H}$, implemented as a lightweight multilayer perceptron with stacked linear transformations and nonlinear activations, transforms this conditioning vector into the *latent code* $g_{s,t} = \mathcal{H}(p_{s,t}) \in \mathbb{R}^{d_h}$. To convert this shared patient–task conditioning into an objective-compatible prediction function, a lightweight *task-specific parameter generator* G_t maps the latent code to the corresponding *dynamic parameters*:

$$[\text{vec}(W_{s,t}); b_{s,t}] = G_t(g_{s,t}), \quad (8)$$

where $W_{s,t} \in \mathbb{R}^{d_a\times o_t}$, $b_{s,t} \in \mathbb{R}^{o_t}$, and o_t denotes the output dimensionality of task t. The final *patient–task-conditioned prediction* is obtained by applying the generated parameters to the patient-aware representation:

$$\hat{y}_{s,t} = f_{s,t}^T W_{s,t} + b_{s,t}. \quad (9)$$

Softmax is applied to multi-class outputs, sigmoid to binary outputs, and continuous outputs are directly regressed. When patient-specific clinical information is unavailable, $h_{s,t}^{\text{cls}} = \mathbf{0}$; parameter generation is then conditioned by task identity alone, while the image-derived representation remains available through the residual visual pathway. The resulting predictor is therefore task-specific but not task-static: its parameters adapt to the available patient context, enabling adaptation across both heterogeneous objectives and patient states within the same objective while retaining a shared prediction architecture.

2.4. Model optimization and inference

The framework is optimized in two sequential stages to separate robust visual feature construction from patient-aware adaptive prediction. In the first stage, the visual encoder is warm-started with task-specific auxiliary heads and subsequently optimized with the clinically induced relation prior: $\mathcal{L}_{\text{stage1}} = \mathcal{L}_{\text{aux}} + \lambda_{\text{prior}}\mathcal{L}_{\text{prior}}$, where $\mathcal{L}_{\text{aux}}$ uses cross-entropy for classification and mean squared error for regression. HSIC- and KCI-based filtering is then performed independently for each task, followed by top-ρ retention to obtain the fixed task-specific channel subsets $\{\mathcal{S}_t\}_{t\in\mathcal{T}}$.

In the second stage, the relation-guided visual encoder and task-specific feature subsets are fixed, while ClinicalBERT, patient-semantic-guided visual contextualization, the shared hypernetwork, and task-specific parameter generators are optimized jointly. For each observed patient–task pair, the available clinical characteristics are processed using the same task-specific masking strategy and clinical context dropout described above. The resulting patient-aware representation is combined with task identity to generate dynamic prediction parameters. The overall objective is written compactly as $\mathcal{L}_{\text{stage2}} = \mathcal{L}_{\text{task}} + \lambda_{\text{hyper}}\mathcal{L}_{\text{hyper}}$. Here, $\mathcal{L}_{\text{task}}$ is the mean of the task-wise prediction losses over the observed task pairs. Standard cross-entropy is used for multi-class diagnosis, binary cross-entropy for binary prognosis, and mean squared error for continuous regression targets. To prevent excessive variation in the dynamically generated prediction parameters, we regularize their magnitude as

$$\mathcal{L}_{\text{hyper}} = \frac{1}{|\mathcal{B}_{\text{img}}|}\sum_{(s,t)\in\mathcal{B}}\left(\|W_{s,t}\|_F^2 + \|b_{s,t}\|_2^2\right), \quad (10)$$

where $\mathcal{B}$ denotes the patient–task pairs in the current mini-batch. This regularization constrains parameter generation while preserving patient–task-specific adaptation.

During inference, a CCM image is first encoded and filtered using the fixed task-specific subset $\mathcal{S}_t$. Available patient information is processed using the same task-specific masking rules and encoded into patient semantics, which contextualize the refined visual representation. The patient semantic embedding is then combined with the requested task identity to condition the shared hypernetwork, and the resulting task-specific dynamic parameters produce the final prediction $\hat{y}_{s,t}$. When patient-specific clinical information is partially or completely unavailable, the same trained framework directly uses the available context; in the fully missing case, inference reduces to the image-derived representation and task identity.

3. Experiments

3.1. Datasets and experimental setup

The study comprised two cross-sectional disease domains and two longitudinal ocular settings, covering diagnosis, clini-

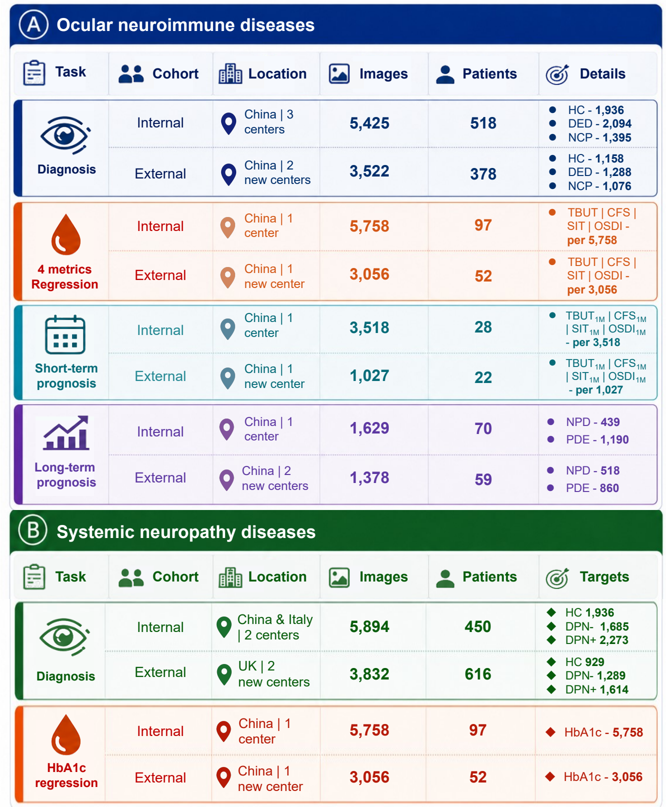

Fig. 3. Overview of datasets and clinical evaluation tasks.

cal reference metric regression, short-term treatment prognosis, and long-term postoperative prognosis. The corresponding cohort composition, participating centers, image and patient numbers, class distributions, and task-specific targets are summarized in Fig. 3. Internal cohorts were used for model development under patient-level cross-validation, whereas independent patients from previously unseen centers were reserved for external evaluation. The ocular-surface reference metric and HbA1c regression tasks were derived from the same regression dataset and therefore share the same image and patient counts.

Ocular neuroimmune assessment. The ocular diagnosis task discriminated HC, DED, and NCP. A separate ocular-surface reference metric regression setting estimated four complementary clinical endpoints—TBUT, CFS, SIT, and OSDI—from CCM images, providing quantitative functional phenotypes in addition to categorical disease discrimination.

Systemic neuropathy assessment. The systemic diagnosis task modeled the progression from HC to T1DM without DPN (DPN−) and T1DM with DPN (DPN+). HbA1c was additionally estimated as a continuous metabolic reference, allowing corneal neural phenotypes and glycemic burden to be evaluated within the same systemic assessment framework.

Short-term treatment prognosis. Baseline (T_0) CCM and clinical measurements were acquired before treatment, followed by reassessment at one month (1M). Given the baseline CCM image and clinical information available at T_0, the model directly predicted the absolute one-month values y_{1M} of TBUT, CFS, SIT, and OSDI. Measurements acquired at 1M were used exclusively as prediction targets and were excluded from model inputs.

Long-term postoperative prognosis. Preoperative baseline (T_0) CCM was used to predict the six-month outcome after FS-LASIK, distinguishing non-persistent dry eye (NPD) from persistent dry eye (PDE). No postoperative information was available to the model at baseline prediction.

Evaluation protocol. All CCM images were acquired using the Heidelberg Retina Tomograph III with the Rostock Cornea Module (HRT-III/RCM) and analyzed at a spatial resolution of 384×384 pixels. External evaluation therefore assessed cross-center generalization under a common imaging platform rather than cross-device generalization. All data partitioning was performed at the patient level: internal cohorts were evaluated using five-fold cross-validation, while external cohorts from independent centers remained fixed and were not involved in model development or hyperparameter selection.

The study was approved by the appropriate institutional ethics committees of the participating centers and by the Institute of Biomedical Engineering, Ningbo Institute of Materials Technology and Engineering, Chinese Academy of Sciences (20240320-0022R). Each center conducted the study under its corresponding local ethical approval or applicable waiver. All images and associated metadata were anonymized before analysis, and all procedures followed the Declaration of Helsinki.

3.2. Implementation details

A ResNet-50 pretrained on ImageNet was used as the visual encoder, and pretrained ClinicalBERT (Tiny) was used for patient-semantic encoding. Continuous clinical variables were standardized using statistics estimated from the corresponding training partition. Training followed the two-stage procedure described in Sec. 2.4. After visual warm-up convergence, Π and M_{prior} were estimated from the training partition and fixed for subsequent relation-guided optimization. AdamW was used with an initial learning rate of 1×10^{-4}, weight decay of 1×10^{-4}, cosine annealing to 1×10^{-6}, and batch size 16. Training was limited to 500 epochs with early stopping after 20 epochs without validation improvement. Based on the hyperparameter sensitivity analysis, λ_{prior}, ρ, and λ_{hyper} were fixed to 1.0, 30%, and 5×10^{-4}, respectively.

All experiments used strict patient-level partitioning. Internal results are reported as mean ± standard deviation across the five held-out folds. For external evaluation, the same fixed cohort was evaluated by the five independently trained fold models, with mean ± standard deviation reflecting variability across models. Repeated predictions corresponding to the same case were aggregated before statistical testing. Statistical comparisons used two-sided paired permutation tests for multi-class AUC and regression metrics and two-sided DeLong tests for binary AUC, with Bonferroni correction for multiple comparisons. All normalization statistics, relation quantities, kernel parameters, and feature-filtering quantities were estimated exclusively from the corresponding training partitions, and external cohorts remained independent until model configuration and hyperparameters had been fixed.

Table 1. Quantitative comparison of ocular neuroimmune assessment on the internal and external cohorts. Reported p-values are calculated against Ours (0% CD). CD denotes clinical context dropout; 0% CD uses complete task-safe patient context, whereas 100% CD denotes image-only inference.

Methods	Diagnosis						Regression					
	Metrics	HC	DED	NCP	Avg.	p <	Metrics	TBUT	CFS	SIT	OSDI	p <
						Internal validation						
JMMLRC	ACC	$0.808_{\pm0.048}$	$0.824_{\pm0.049}$	$0.846_{\pm0.032}$	$0.826_{\pm0.039}$		RMSE	$7.247_{\pm0.234}$	$7.104_{\pm0.306}$	$7.481_{\pm0.295}$	$7.168_{\pm0.364}$	
	AUC	$0.848_{\pm0.031}$	$0.862_{\pm0.027}$	$0.887_{\pm0.030}$	$0.866_{\pm0.021}$	.001	MAE	$5.233_{\pm0.266}$	$5.116_{\pm0.137}$	$5.469_{\pm0.283}$	$5.082_{\pm0.156}$	.001
QCNN	ACC	$0.839_{\pm0.022}$	$0.838_{\pm0.027}$	$0.852_{\pm0.040}$	$0.843_{\pm0.024}$		RMSE	$6.716_{\pm0.297}$	$6.381_{\pm0.213}$	$7.126_{\pm0.266}$	$6.549_{\pm0.346}$	
	AUC	$0.879_{\pm0.023}$	$0.903_{\pm0.021}$	$0.883_{\pm0.018}$	$0.888_{\pm0.014}$	.001	MAE	$4.524_{\pm0.136}$	$4.097_{\pm0.224}$	$4.834_{\pm0.267}$	$4.212_{\pm0.144}$	.001
C-former	ACC	$0.826_{\pm0.031}$	$0.831_{\pm0.046}$	$0.885_{\pm0.030}$	$0.847_{\pm0.028}$		RMSE	$5.846_{\pm0.267}$	$5.723_{\pm0.155}$	$6.357_{\pm0.336}$	$6.874_{\pm0.221}$	
	AUC	$0.872_{\pm0.027}$	$0.886_{\pm0.022}$	$0.920_{\pm0.019}$	$0.893_{\pm0.021}$	.001	MAE	$3.414_{\pm0.114}$	$3.187_{\pm0.203}$	$3.923_{\pm0.136}$	$4.136_{\pm0.227}$	.001
Multi-Cog	ACC	$0.852_{\pm0.028}$	$0.854_{\pm0.031}$	$0.868_{\pm0.027}$	$0.858_{\pm0.020}$		RMSE	$6.129_{\pm0.184}$	$5.751_{\pm0.276}$	$6.596_{\pm0.254}$	$6.012_{\pm0.333}$	
	AUC	$0.894_{\pm0.011}$	$0.915_{\pm0.018}$	$0.897_{\pm0.022}$	$0.902_{\pm0.014}$	.001	MAE	$3.867_{\pm0.226}$	$3.443_{\pm0.117}$	$4.114_{\pm0.246}$	$3.576_{\pm0.131}$	.001
MM-MIL	ACC	$0.817_{\pm0.043}$	$0.815_{\pm0.052}$	$0.859_{\pm0.022}$	$0.830_{\pm0.031}$		RMSE	$6.984_{\pm0.315}$	$6.642_{\pm0.195}$	$7.318_{\pm0.386}$	$6.906_{\pm0.255}$	
	AUC	$0.859_{\pm0.018}$	$0.875_{\pm0.021}$	$0.904_{\pm0.023}$	$0.879_{\pm0.016}$	.001	MAE	$4.896_{\pm0.153}$	$4.423_{\pm0.233}$	$5.084_{\pm0.176}$	$4.653_{\pm0.245}$	.001
MM-DLS	ACC	$0.834_{\pm0.037}$	$0.842_{\pm0.041}$	$0.873_{\pm0.019}$	$0.850_{\pm0.022}$		RMSE	$5.857_{\pm0.176}$	$5.594_{\pm0.267}$	$6.371_{\pm0.244}$	$5.786_{\pm0.316}$	
	AUC	$0.887_{\pm0.012}$	$0.896_{\pm0.010}$	$0.891_{\pm0.027}$	$0.891_{\pm0.013}$	.001	MAE	$3.573_{\pm0.217}$	$3.236_{\pm0.106}$	$3.883_{\pm0.243}$	$3.347_{\pm0.124}$	.001
DenseMoE	ACC	$0.844_{\pm0.019}$	$0.856_{\pm0.023}$	$0.873_{\pm0.031}$	$0.858_{\pm0.019}$		RMSE	$5.636_{\pm0.264}$	$5.243_{\pm0.144}$	$7.164_{\pm0.327}$	$7.513_{\pm0.216}$	
	AUC	$0.895_{\pm0.018}$	$0.905_{\pm0.019}$	$0.902_{\pm0.012}$	$0.901_{\pm0.013}$	.001	MAE	$2.987_{\pm0.206}$	$3.064_{\pm0.194}$	$3.756_{\pm0.135}$	$3.127_{\pm0.223}$	.001
KGT	ACC	$0.847_{\pm0.030}$	$0.859_{\pm0.031}$	$0.887_{\pm0.017}$	$0.864_{\pm0.018}$		RMSE	$5.828_{\pm0.283}$	$5.013_{\pm0.158}$	$7.625_{\pm0.361}$	$7.247_{\pm0.244}$	
	AUC	$0.897_{\pm0.019}$	$0.910_{\pm0.016}$	$0.916_{\pm0.018}$	$0.908_{\pm0.014}$	.001	MAE	$2.746_{\pm0.118}$	$3.154_{\pm0.202}$	$3.621_{\pm0.141}$	$2.889_{\pm0.232}$	.001
Ours (100% CD)	ACC	$0.872_{\pm0.043}$	$0.902_{\pm0.029}$	$0.917_{\pm0.024}$	$0.897_{\pm0.025}$		RMSE	$2.438_{\pm0.226}$	$2.285_{\pm0.108}$	$4.173_{\pm0.257}$	$4.526_{\pm0.148}$	
	AUC	$0.917_{\pm0.018}$	$0.931_{\pm0.021}$	$0.942_{\pm0.026}$	$0.930_{\pm0.015}$	.001	MAE	$1.741_{\pm0.184}$	$1.439_{\pm0.167}$	$2.164_{\pm0.098}$	$1.983_{\pm0.186}$	.001
Ours (0% CD)	**ACC**	$0.901_{\pm0.032}$	$0.914_{\pm0.033}$	$0.940_{\pm0.019}$	$0.918_{\pm0.020}$		**RMSE**	$1.699_{\pm0.197}$	$1.574_{\pm0.083}$	$3.564_{\pm0.234}$	$3.773_{\pm0.122}$	
	AUC	$0.938_{\pm0.018}$	$0.950_{\pm0.011}$	$0.963_{\pm0.027}$	$0.950_{\pm0.015}$	–	**MAE**	$1.092_{\pm0.064}$	$0.815_{\pm0.146}$	$1.500_{\pm0.077}$	$1.348_{\pm0.163}$	–
						External validation						
Multi-Cog	ACC	$0.781_{\pm0.049}$	$0.826_{\pm0.040}$	$0.813_{\pm0.035}$	$0.807_{\pm0.032}$		RMSE	$7.491_{\pm0.265}$	$6.862_{\pm0.342}$	$7.683_{\pm0.385}$	$7.128_{\pm0.426}$	
	AUC	$0.803_{\pm0.031}$	$0.805_{\pm0.028}$	$0.854_{\pm0.027}$	$0.821_{\pm0.023}$	.001	MAE	$5.274_{\pm0.292}$	$5.148_{\pm0.161}$	$5.345_{\pm0.314}$	$5.267_{\pm0.169}$	.001
JMMLRC	ACC	$0.802_{\pm0.030}$	$0.847_{\pm0.024}$	$0.839_{\pm0.036}$	$0.829_{\pm0.024}$		RMSE	$6.762_{\pm0.333}$	$6.427_{\pm0.196}$	$6.985_{\pm0.441}$	$6.758_{\pm0.289}$	
	AUC	$0.837_{\pm0.031}$	$0.835_{\pm0.028}$	$0.847_{\pm0.020}$	$0.840_{\pm0.019}$	.001	MAE	$4.789_{\pm0.160}$	$4.214_{\pm0.231}$	$4.962_{\pm0.185}$	$4.463_{\pm0.253}$	.001
MM-DLS	ACC	$0.793_{\pm0.036}$	$0.820_{\pm0.041}$	$0.875_{\pm0.016}$	$0.829_{\pm0.025}$		RMSE	$6.038_{\pm0.190}$	$5.619_{\pm0.283}$	$6.542_{\pm0.304}$	$6.953_{\pm0.376}$	
	AUC	$0.825_{\pm0.036}$	$0.841_{\pm0.015}$	$0.868_{\pm0.023}$	$0.845_{\pm0.017}$	.001	MAE	$3.571_{\pm0.231}$	$3.385_{\pm0.108}$	$4.027_{\pm0.252}$	$4.456_{\pm0.137}$	.001
C-former	ACC	$0.819_{\pm0.035}$	$0.838_{\pm0.026}$	$0.862_{\pm0.029}$	$0.840_{\pm0.022}$		RMSE	$6.145_{\pm0.319}$	$5.746_{\pm0.178}$	$6.418_{\pm0.423}$	$6.172_{\pm0.268}$	
	AUC	$0.848_{\pm0.025}$	$0.846_{\pm0.027}$	$0.864_{\pm0.014}$	$0.853_{\pm0.017}$	.001	MAE	$4.121_{\pm0.138}$	$3.529_{\pm0.229}$	$4.293_{\pm0.159}$	$3.762_{\pm0.248}$	.001
QCNN	ACC	$0.784_{\pm0.037}$	$0.811_{\pm0.032}$	$0.828_{\pm0.041}$	$0.808_{\pm0.029}$		RMSE	$7.174_{\pm0.241}$	$6.558_{\pm0.324}$	$7.419_{\pm0.353}$	$6.912_{\pm0.422}$	
	AUC	$0.814_{\pm0.033}$	$0.834_{\pm0.029}$	$0.844_{\pm0.021}$	$0.831_{\pm0.021}$	.001	MAE	$4.962_{\pm0.278}$	$4.685_{\pm0.142}$	$5.118_{\pm0.301}$	$4.921_{\pm0.164}$	.001
KGT	ACC	$0.796_{\pm0.043}$	$0.843_{\pm0.033}$	$0.854_{\pm0.024}$	$0.831_{\pm0.025}$		RMSE	$6.056_{\pm0.298}$	$5.443_{\pm0.189}$	$6.364_{\pm0.398}$	$5.941_{\pm0.262}$	
	AUC	$0.855_{\pm0.012}$	$0.838_{\pm0.016}$	$0.857_{\pm0.026}$	$0.850_{\pm0.014}$	.001	MAE	$3.727_{\pm0.141}$	$3.446_{\pm0.211}$	$3.948_{\pm0.164}$	$3.675_{\pm0.232}$	.001
DenseMoE	ACC	$0.819_{\pm0.024}$	$0.847_{\pm0.020}$	$0.849_{\pm0.030}$	$0.838_{\pm0.020}$		RMSE	$5.858_{\pm0.182}$	$4.894_{\pm0.271}$	$7.305_{\pm0.261}$	$7.642_{\pm0.364}$	
	AUC	$0.833_{\pm0.028}$	$0.842_{\pm0.024}$	$0.865_{\pm0.010}$	$0.847_{\pm0.013}$	.001	MAE	$3.126_{\pm0.222}$	$3.392_{\pm0.100}$	$3.381_{\pm0.243}$	$3.052_{\pm0.124}$	.001
MM-MIL	ACC	$0.816_{\pm0.027}$	$0.852_{\pm0.034}$	$0.860_{\pm0.020}$	$0.843_{\pm0.021}$		RMSE	$5.971_{\pm0.204}$	$4.682_{\pm0.268}$	$7.593_{\pm0.282}$	$7.424_{\pm0.401}$	
	AUC	$0.853_{\pm0.027}$	$0.855_{\pm0.014}$	$0.869_{\pm0.021}$	$0.859_{\pm0.013}$	.001	MAE	$2.869_{\pm0.232}$	$3.041_{\pm0.115}$	$2.858_{\pm0.251}$	$3.393_{\pm0.136}$	.001
Ours (100% CD)	ACC	$0.856_{\pm0.023}$	$0.888_{\pm0.016}$	$0.893_{\pm0.039}$	$0.879_{\pm0.021}$		RMSE	$2.486_{\pm0.219}$	$1.872_{\pm0.083}$	$4.219_{\pm0.244}$	$4.683_{\pm0.161}$	
	AUC	$0.872_{\pm0.025}$	$0.866_{\pm0.040}$	$0.894_{\pm0.011}$	$0.877_{\pm0.020}$	.001	MAE	$1.815_{\pm0.088}$	$1.364_{\pm0.162}$	$1.937_{\pm0.086}$	$1.834_{\pm0.171}$	.001
Ours (0% CD)	**ACC**	$0.871_{\pm0.028}$	$0.906_{\pm0.024}$	$0.922_{\pm0.031}$	$0.900_{\pm0.022}$		**RMSE**	$1.810_{\pm0.102}$	$1.145_{\pm0.167}$	$3.428_{\pm0.128}$	$3.958_{\pm0.243}$	
	AUC	$0.898_{\pm0.032}$	$0.899_{\pm0.029}$	$0.920_{\pm0.016}$	$0.906_{\pm0.016}$	–	**MAE**	$1.232_{\pm0.173}$	$0.829_{\pm0.043}$	$1.206_{\pm0.168}$	$1.117_{\pm0.054}$	–

3.3. Evaluation metrics

Classification performance was evaluated using class-wise accuracy (ACC) and area under the receiver operating characteristic curve (AUC). Class-wise ACC was calculated as the proportion of correctly classified samples within each class, with the average obtained by macro-averaging across classes. For multi-class tasks, AUC was computed in a one-vs-rest manner and macro-averaged across classes. Regression performance was assessed using root mean square error (RMSE) and mean absolute error (MAE). Higher ACC/AUC and lower RMSE/MAE indicate better performance.

3.4. Comparison methods

We compared our framework with eight representative medical multi-task methods capable of jointly performing diagnostic classification and continuous regression: JMMLRC Brand et al. (2020), QCNN (Lao et al., 2025), C-former (Mineo et al., 2024), Multi-Cog (Ma et al., 2026), MM-MIL (Huang et al., 2026), MM-DLS (Liu et al., 2026), DenseMoE (Ding et al., 2025), and KGT Zhao et al. (2026). All comparison methods were evaluated using the same data partitions and task-specific experimental settings as the proposed framework to ensure a consistent comparison.

Table 2. Quantitative comparison of systemic neuropathy assessment on the internal and external cohorts. Reported p-values are calculated against Ours (0% CD). CD denotes clinical context dropout; 0% CD uses complete task-safe patient context, whereas 100% CD denotes image-only inference.

Methods	Diagnosis						Regression	
	Metrics	HC	DPN-	DPN+	Avg.	p <	Metrics	HbA1c
Internal validation								
JMMLRC	ACC	$0.814_{\pm0.041}$	$0.783_{\pm0.037}$	$0.846_{\pm0.034}$	$0.814_{\pm0.028}$		RMSE	$7.286_{\pm0.246}$
	AUC	$0.858_{\pm0.013}$	$0.834_{\pm0.028}$	$0.872_{\pm0.011}$	$0.855_{\pm0.014}$	.001	MAE	$5.274_{\pm0.259}$
QCNN	ACC	$0.840_{\pm0.033}$	$0.797_{\pm0.032}$	$0.858_{\pm0.030}$	$0.832_{\pm0.022}$		RMSE	$6.733_{\pm0.317}$
	AUC	$0.880_{\pm0.019}$	$0.851_{\pm0.024}$	$0.887_{\pm0.008}$	$0.873_{\pm0.013}$	.001	MAE	$4.614_{\pm0.136}$
C-former	ACC	$0.831_{\pm0.028}$	$0.825_{\pm0.034}$	$0.863_{\pm0.021}$	$0.840_{\pm0.022}$		RMSE	$5.974_{\pm0.192}$
	AUC	$0.872_{\pm0.022}$	$0.884_{\pm0.006}$	$0.880_{\pm0.022}$	$0.879_{\pm0.013}$	.001	MAE	$3.453_{\pm0.213}$
Multi-Cog	ACC	$0.853_{\pm0.018}$	$0.806_{\pm0.038}$	$0.874_{\pm0.024}$	$0.844_{\pm0.021}$		RMSE	$6.117_{\pm0.306}$
	AUC	$0.892_{\pm0.016}$	$0.864_{\pm0.010}$	$0.899_{\pm0.007}$	$0.885_{\pm0.009}$	.001	MAE	$3.896_{\pm0.127}$
MM-MIL	ACC	$0.821_{\pm0.039}$	$0.790_{\pm0.034}$	$0.852_{\pm0.033}$	$0.821_{\pm0.026}$		RMSE	$6.976_{\pm0.231}$
	AUC	$0.867_{\pm0.011}$	$0.845_{\pm0.024}$	$0.881_{\pm0.010}$	$0.864_{\pm0.012}$	.001	MAE	$4.947_{\pm0.247}$
MM-DLS	ACC	$0.837_{\pm0.025}$	$0.814_{\pm0.036}$	$0.867_{\pm0.018}$	$0.839_{\pm0.021}$		RMSE	$5.914_{\pm0.299}$
	AUC	$0.884_{\pm0.018}$	$0.860_{\pm0.011}$	$0.893_{\pm0.019}$	$0.879_{\pm0.013}$	.001	MAE	$3.586_{\pm0.122}$
DenseMoE	ACC	$0.851_{\pm0.026}$	$0.809_{\pm0.024}$	$0.872_{\pm0.023}$	$0.844_{\pm0.016}$		RMSE	$4.996_{\pm0.193}$
	AUC	$0.888_{\pm0.005}$	$0.867_{\pm0.018}$	$0.895_{\pm0.006}$	$0.883_{\pm0.008}$	.001	MAE	$2.853_{\pm0.214}$
KGT	ACC	$0.857_{\pm0.018}$	$0.821_{\pm0.032}$	$0.883_{\pm0.024}$	$0.854_{\pm0.019}$		RMSE	$4.832_{\pm0.285}$
	AUC	$0.890_{\pm0.016}$	$0.888_{\pm0.005}$	$0.897_{\pm0.018}$	$0.892_{\pm0.005}$	.001	MAE	$2.571_{\pm0.109}$
Ours (100% CD)	ACC	$0.902_{\pm0.028}$	$0.850_{\pm0.023}$	$0.921_{\pm0.032}$	$0.891_{\pm0.022}$		RMSE	$2.148_{\pm0.196}$
	AUC	$0.931_{\pm0.016}$	$0.904_{\pm0.024}$	$0.934_{\pm0.019}$	$0.923_{\pm0.013}$	.001	MAE	$1.439_{\pm0.154}$
Ours (0% CD)	**ACC**	$0.912_{\pm0.020}$	$0.882_{\pm0.034}$	$0.934_{\pm0.016}$	$0.909_{\pm0.018}$		**RMSE**	$1.276_{\pm0.072}$
	AUC	$0.938_{\pm0.021}$	$0.929_{\pm0.008}$	$0.946_{\pm0.022}$	$0.938_{\pm0.013}$	–	**MAE**	$0.820_{\pm0.137}$
External validation								
C-former	ACC	$0.795_{\pm0.045}$	$0.790_{\pm0.037}$	$0.826_{\pm0.038}$	$0.804_{\pm0.030}$		RMSE	$7.428_{\pm0.259}$
	AUC	$0.844_{\pm0.030}$	$0.814_{\pm0.028}$	$0.818_{\pm0.027}$	$0.825_{\pm0.022}$	.001	MAE	$5.286_{\pm0.279}$
MM-MIL	ACC	$0.835_{\pm0.027}$	$0.818_{\pm0.035}$	$0.821_{\pm0.029}$	$0.825_{\pm0.024}$		RMSE	$6.784_{\pm0.339}$
	AUC	$0.850_{\pm0.034}$	$0.831_{\pm0.017}$	$0.843_{\pm0.029}$	$0.841_{\pm0.019}$	.001	MAE	$4.815_{\pm0.148}$
QCNN	ACC	$0.856_{\pm0.041}$	$0.789_{\pm0.027}$	$0.832_{\pm0.027}$	$0.826_{\pm0.025}$		RMSE	$5.823_{\pm0.308}$
	AUC	$0.862_{\pm0.025}$	$0.850_{\pm0.015}$	$0.859_{\pm0.024}$	$0.857_{\pm0.014}$	.001	MAE	$3.662_{\pm0.135}$
KGT	ACC	$0.814_{\pm0.034}$	$0.806_{\pm0.041}$	$0.870_{\pm0.024}$	$0.830_{\pm0.023}$		RMSE	$6.191_{\pm0.230}$
	AUC	$0.871_{\pm0.021}$	$0.846_{\pm0.020}$	$0.875_{\pm0.019}$	$0.864_{\pm0.016}$	.001	MAE	$3.979_{\pm0.236}$
JMMLRC	ACC	$0.805_{\pm0.042}$	$0.799_{\pm0.038}$	$0.830_{\pm0.025}$	$0.811_{\pm0.028}$		RMSE	$7.126_{\pm0.345}$
	AUC	$0.852_{\pm0.027}$	$0.820_{\pm0.024}$	$0.822_{\pm0.038}$	$0.831_{\pm0.022}$	.001	MAE	$5.038_{\pm0.168}$
DenseMoE	ACC	$0.843_{\pm0.035}$	$0.822_{\pm0.020}$	$0.839_{\pm0.026}$	$0.835_{\pm0.021}$		RMSE	$5.989_{\pm0.224}$
	AUC	$0.857_{\pm0.021}$	$0.839_{\pm0.026}$	$0.855_{\pm0.026}$	$0.850_{\pm0.017}$	.001	MAE	$3.662_{\pm0.244}$
MM-DLS	ACC	$0.831_{\pm0.031}$	$0.820_{\pm0.025}$	$0.843_{\pm0.030}$	$0.831_{\pm0.021}$		RMSE	$5.374_{\pm0.301}$
	AUC	$0.869_{\pm0.016}$	$0.848_{\pm0.018}$	$0.857_{\pm0.013}$	$0.858_{\pm0.012}$	.001	MAE	$3.147_{\pm0.126}$
Multi-Cog	ACC	$0.842_{\pm0.026}$	$0.825_{\pm0.029}$	$0.860_{\pm0.017}$	$0.842_{\pm0.019}$		RMSE	$4.949_{\pm0.199}$
	AUC	$0.880_{\pm0.024}$	$0.853_{\pm0.019}$	$0.871_{\pm0.022}$	$0.868_{\pm0.010}$	.001	MAE	$3.092_{\pm0.225}$
Ours (100% CD)	ACC	$0.890_{\pm0.021}$	$0.855_{\pm0.034}$	$0.894_{\pm0.020}$	$0.880_{\pm0.018}$		RMSE	$2.046_{\pm0.197}$
	AUC	$0.901_{\pm0.029}$	$0.883_{\pm0.017}$	$0.892_{\pm0.026}$	$0.892_{\pm0.019}$	.001	MAE	$1.618_{\pm0.058}$
Ours (0% CD)	**ACC**	$0.907_{\pm0.026}$	$0.876_{\pm0.032}$	$0.912_{\pm0.019}$	$0.898_{\pm0.020}$		**RMSE**	$1.385_{\pm0.082}$
	AUC	$0.923_{\pm0.031}$	$0.902_{\pm0.014}$	$0.915_{\pm0.028}$	$0.913_{\pm0.016}$	–	**MAE**	$0.965_{\pm0.142}$

Table 3. Quantitative comparison of short-term treatment prognosis and long-term postoperative prognosis on the internal and external cohorts. Reported p-values are calculated against Ours (0% CD). CD denotes clinical context dropout; 0% CD uses complete task-safe patient context, whereas 100% CD denotes image-only inference.

Methods	Short-term treatment prognosis						Long-term postoperative prognosis				
	Metrics	$TBUT_{1M}$	CFS_{1M}	SIT_{1M}	$OSDI_{1M}$	p <	Metrics	Non-Pers.	Pers.	Avg.	p <
Internal validation											
C-former	RMSE	$7.383_{\pm0.266}$	$7.123_{\pm0.333}$	$7.534_{\pm0.357}$	$7.282_{\pm0.404}$		ACC	$0.834_{\pm0.040}$	$0.806_{\pm0.041}$	$0.820_{\pm0.042}$	
	MAE	$5.269_{\pm0.274}$	$5.114_{\pm0.145}$	$5.408_{\pm0.302}$	$5.029_{\pm0.186}$	.001	AUC	$0.893_{\pm0.016}$	$0.893_{\pm0.016}$	$0.893_{\pm0.016}$	.001
MM-DLS	RMSE	$6.876_{\pm0.241}$	$6.565_{\pm0.314}$	$7.168_{\pm0.312}$	$6.726_{\pm0.389}$		ACC	$0.852_{\pm0.034}$	$0.817_{\pm0.037}$	$0.835_{\pm0.038}$	
	MAE	$4.691_{\pm0.248}$	$4.248_{\pm0.136}$	$4.936_{\pm0.283}$	$4.366_{\pm0.165}$	.001	AUC	$0.916_{\pm0.019}$	$0.916_{\pm0.019}$	$0.916_{\pm0.019}$	.001
JMMLRC	RMSE	$6.315_{\pm0.305}$	$6.107_{\pm0.196}$	$6.814_{\pm0.392}$	$7.263_{\pm0.268}$		ACC	$0.843_{\pm0.029}$	$0.846_{\pm0.035}$	$0.845_{\pm0.023}$	
	MAE	$3.639_{\pm0.124}$	$3.386_{\pm0.213}$	$4.183_{\pm0.167}$	$4.419_{\pm0.244}$	.001	AUC	$0.908_{\pm0.023}$	$0.908_{\pm0.023}$	$0.908_{\pm0.023}$	.001
DenseMoE	RMSE	$6.364_{\pm0.232}$	$6.044_{\pm0.309}$	$6.649_{\pm0.295}$	$6.201_{\pm0.370}$		ACC	$0.865_{\pm0.030}$	$0.824_{\pm0.034}$	$0.845_{\pm0.036}$	
	MAE	$4.085_{\pm0.237}$	$3.631_{\pm0.126}$	$4.319_{\pm0.273}$	$3.758_{\pm0.156}$	.001	AUC	$0.923_{\pm0.007}$	$0.923_{\pm0.007}$	$0.923_{\pm0.007}$	.001
QCNN	RMSE	$7.107_{\pm0.256}$	$6.890_{\pm0.328}$	$7.395_{\pm0.333}$	$6.954_{\pm0.401}$		ACC	$0.827_{\pm0.044}$	$0.819_{\pm0.036}$	$0.823_{\pm0.039}$	
	MAE	$4.934_{\pm0.261}$	$4.571_{\pm0.139}$	$5.177_{\pm0.295}$	$4.697_{\pm0.168}$	.001	AUC	$0.901_{\pm0.012}$	$0.901_{\pm0.012}$	$0.901_{\pm0.012}$	.001
Multi-Cog	RMSE	$6.143_{\pm0.319}$	$5.932_{\pm0.204}$	$6.436_{\pm0.406}$	$6.093_{\pm0.277}$		ACC	$0.846_{\pm0.028}$	$0.837_{\pm0.039}$	$0.842_{\pm0.024}$	
	MAE	$3.764_{\pm0.131}$	$3.310_{\pm0.220}$	$4.011_{\pm0.164}$	$3.530_{\pm0.248}$	.001	AUC	$0.912_{\pm0.021}$	$0.912_{\pm0.021}$	$0.912_{\pm0.021}$	.001
KGT	RMSE	$6.217_{\pm0.297}$	$5.505_{\pm0.183}$	$6.903_{\pm0.361}$	$7.364_{\pm0.246}$		ACC	$0.866_{\pm0.026}$	$0.834_{\pm0.028}$	$0.850_{\pm0.029}$	
	MAE	$3.167_{\pm0.115}$	$3.400_{\pm0.208}$	$3.805_{\pm0.142}$	$3.024_{\pm0.223}$	.001	AUC	$0.923_{\pm0.013}$	$0.923_{\pm0.013}$	$0.923_{\pm0.013}$	.001
MM-MIL	RMSE	$6.124_{\pm0.308}$	$5.808_{\pm0.185}$	$7.132_{\pm0.269}$	$6.867_{\pm0.354}$		ACC	$0.871_{\pm0.019}$	$0.843_{\pm0.033}$	$0.857_{\pm0.017}$	
	MAE	$2.847_{\pm0.121}$	$3.239_{\pm0.206}$	$4.085_{\pm0.243}$	$2.724_{\pm0.135}$	.001	AUC	$0.929_{\pm0.014}$	$0.929_{\pm0.014}$	$0.929_{\pm0.014}$	.001
Ours (100% CD)	RMSE	$3.247_{\pm0.229}$	$3.186_{\pm0.112}$	$4.392_{\pm0.278}$	$4.639_{\pm0.164}$		ACC	$0.904_{\pm0.021}$	$0.880_{\pm0.036}$	$0.892_{\pm0.040}$	
	MAE	$1.938_{\pm0.084}$	$1.625_{\pm0.168}$	$2.483_{\pm0.102}$	$1.812_{\pm0.174}$	.001	AUC	$0.958_{\pm0.028}$	$0.958_{\pm0.028}$	$0.958_{\pm0.028}$	.001
Ours (0% CD)	**RMSE**	$2.472_{\pm0.112}$	$2.352_{\pm0.196}$	$3.643_{\pm0.163}$	$3.714_{\pm0.248}$		**ACC**	$0.923_{\pm0.032}$	$0.904_{\pm0.026}$	$0.914_{\pm0.030}$	
	MAE	$1.199_{\pm0.167}$	$1.043_{\pm0.045}$	$1.827_{\pm0.184}$	$1.166_{\pm0.056}$	–	**AUC**	$0.973_{\pm0.015}$	$0.973_{\pm0.015}$	$0.973_{\pm0.015}$	–
External validation											
C-former	RMSE	$7.476_{\pm0.395}$	$7.260_{\pm0.262}$	$7.617_{\pm0.466}$	$7.387_{\pm0.324}$		ACC	$0.884_{\pm0.023}$	$0.758_{\pm0.057}$	$0.821_{\pm0.028}$	
	MAE	$5.416_{\pm0.188}$	$5.175_{\pm0.269}$	$5.360_{\pm0.203}$	$5.091_{\pm0.297}$	.001	AUC	$0.826_{\pm0.042}$	$0.826_{\pm0.042}$	$0.826_{\pm0.042}$	.001
DenseMoE	RMSE	$6.913_{\pm0.359}$	$6.605_{\pm0.226}$	$7.156_{\pm0.443}$	$6.736_{\pm0.294}$		ACC	$0.876_{\pm0.037}$	$0.816_{\pm0.032}$	$0.846_{\pm0.035}$	
	MAE	$4.753_{\pm0.164}$	$4.313_{\pm0.241}$	$4.814_{\pm0.178}$	$4.434_{\pm0.266}$	.001	AUC	$0.853_{\pm0.018}$	$0.853_{\pm0.018}$	$0.853_{\pm0.018}$	.001
MM-MIL	RMSE	$6.165_{\pm0.220}$	$5.857_{\pm0.298}$	$6.705_{\pm0.297}$	$7.183_{\pm0.373}$		ACC	$0.918_{\pm0.015}$	$0.786_{\pm0.048}$	$0.852_{\pm0.019}$	
	MAE	$3.680_{\pm0.235}$	$3.436_{\pm0.126}$	$4.139_{\pm0.254}$	$4.369_{\pm0.147}$	.001	AUC	$0.857_{\pm0.030}$	$0.857_{\pm0.030}$	$0.857_{\pm0.030}$	.001
KGT	RMSE	$6.408_{\pm0.337}$	$6.092_{\pm0.212}$	$6.646_{\pm0.419}$	$6.219_{\pm0.285}$		ACC	$0.895_{\pm0.031}$	$0.808_{\pm0.030}$	$0.852_{\pm0.030}$	
	MAE	$4.126_{\pm0.155}$	$3.685_{\pm0.234}$	$4.391_{\pm0.170}$	$3.803_{\pm0.258}$	.001	AUC	$0.864_{\pm0.016}$	$0.864_{\pm0.016}$	$0.864_{\pm0.016}$	.001
MM-DLS	RMSE	$7.157_{\pm0.263}$	$6.846_{\pm0.335}$	$7.390_{\pm0.351}$	$6.978_{\pm0.406}$		ACC	$0.889_{\pm0.030}$	$0.769_{\pm0.045}$	$0.829_{\pm0.026}$	
	MAE	$4.989_{\pm0.276}$	$4.648_{\pm0.159}$	$5.035_{\pm0.294}$	$4.752_{\pm0.172}$	.001	AUC	$0.835_{\pm0.029}$	$0.835_{\pm0.029}$	$0.835_{\pm0.029}$	.001
JMMLRC	RMSE	$6.194_{\pm0.230}$	$5.882_{\pm0.307}$	$6.429_{\pm0.311}$	$6.015_{\pm0.376}$		ACC	$0.912_{\pm0.024}$	$0.783_{\pm0.039}$	$0.848_{\pm0.019}$	
	MAE	$3.817_{\pm0.238}$	$3.374_{\pm0.127}$	$4.066_{\pm0.271}$	$3.587_{\pm0.143}$	.001	AUC	$0.869_{\pm0.012}$	$0.869_{\pm0.012}$	$0.869_{\pm0.012}$	.001
Multi-Cog	RMSE	$6.069_{\pm0.314}$	$5.549_{\pm0.181}$	$7.004_{\pm0.378}$	$7.472_{\pm0.256}$		ACC	$0.901_{\pm0.016}$	$0.812_{\pm0.036}$	$0.857_{\pm0.017}$	
	MAE	$3.494_{\pm0.132}$	$3.651_{\pm0.214}$	$3.541_{\pm0.147}$	$3.070_{\pm0.229}$	.001	AUC	$0.866_{\pm0.020}$	$0.866_{\pm0.020}$	$0.866_{\pm0.020}$	.001
QCNN	RMSE	$6.783_{\pm0.216}$	$6.047_{\pm0.302}$	$6.826_{\pm0.271}$	$8.239_{\pm0.403}$		ACC	$0.923_{\pm0.021}$	$0.814_{\pm0.027}$	$0.869_{\pm0.024}$	
	MAE	$3.349_{\pm0.233}$	$3.971_{\pm0.119}$	$3.283_{\pm0.205}$	$2.802_{\pm0.117}$	.001	AUC	$0.872_{\pm0.013}$	$0.872_{\pm0.013}$	$0.872_{\pm0.013}$	.001
Ours (100% CD)	RMSE	$3.315_{\pm0.231}$	$3.436_{\pm0.124}$	$4.127_{\pm0.263}$	$4.839_{\pm0.184}$		ACC	$0.957_{\pm0.026}$	$0.841_{\pm0.027}$	$0.899_{\pm0.033}$	
	MAE	$2.584_{\pm0.096}$	$2.315_{\pm0.188}$	$1.762_{\pm0.069}$	$1.815_{\pm0.165}$	.001	AUC	$0.898_{\pm0.019}$	$0.898_{\pm0.019}$	$0.898_{\pm0.019}$	.001
Ours (0% CD)	**RMSE**	$2.583_{\pm0.115}$	$2.678_{\pm0.206}$	$3.418_{\pm0.146}$	$3.979_{\pm0.265}$		**ACC**	$0.974_{\pm0.029}$	$0.865_{\pm0.031}$	$0.920_{\pm0.021}$	
	MAE	$1.805_{\pm0.177}$	$1.770_{\pm0.069}$	$1.076_{\pm0.143}$	$1.176_{\pm0.047}$	–	**AUC**	$0.916_{\pm0.017}$	$0.916_{\pm0.017}$	$0.916_{\pm0.017}$	–

4. Experimental Results

4.1. Quantitative comparison

Tables 1–3 compare the proposed framework with representative multi-task methods across six clinical objectives on both internal and independent external cohorts. Overall, Ours achieved the best performance across diagnosis, continuous regression, and longitudinal prognosis, with the advantage consistently preserved under external validation.

For ocular neuroimmune diagnosis, Ours improved the average AUC over the strongest competing method by 0.042 internally (0.950 vs. 0.908) and 0.047 externally (0.906 vs. 0.859). The advantage was particularly pronounced for continuous prediction: ocular-surface reference metric regression MAE decreased from 3.103 to 1.189 internally and from 3.040 to 1.096 externally, while HbA1c regression decreased from 2.571 to 0.820 and from 3.092 to 0.965, respectively. Similar improvements were observed for longitudinal prediction, with long-term prognosis AUC increasing by 0.044 on both the internal (0.973 vs. 0.929) and external (0.916 vs. 0.872) cohorts; Tables 1–3 provide the complete task-wise results.

The consistent margins across categorical, continuous, and longitudinal outcomes indicate that the proposed framework provides stable multi-task modeling rather than gains confined to a particular endpoint. Importantly, the retained advantages on independent external cohorts further support robust cross-center generalization, while the larger improvements in continuous and prognostic targets suggest more reliable modeling of fine-grained and temporally varying clinical phenotypes.

4.2. Qualitative analysis

Fig. 4 qualitatively evaluates whether the proposed framework focuses on clinically meaningful CCM patterns and whether the learned representations are consistent with established neuro-morphometric changes. Overall, our method shows more concentrated attention on disease-relevant nerve

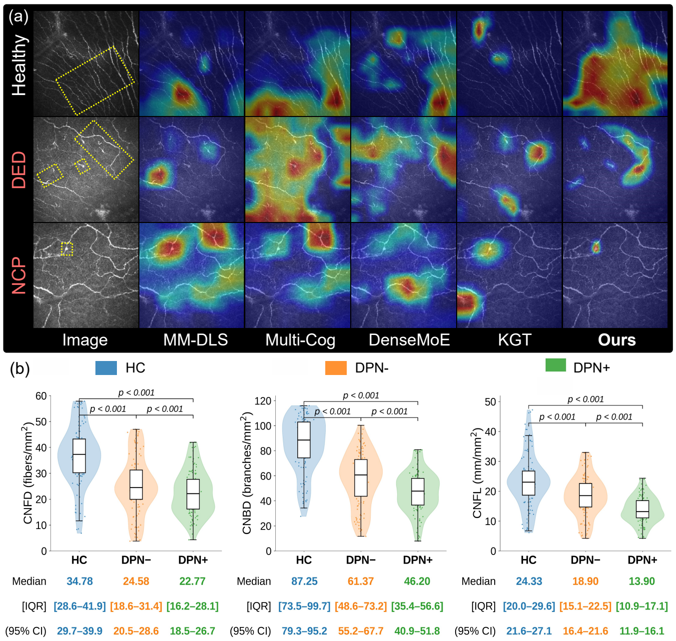

Fig. 4. Qualitative comparison and neuro-morphometric analysis. (a) Grad-CAM visualizations across representative CCM phenotypes. (b) Distributions of CNFD, CNBD, and CNFL across HC, DPN−, and DPN+.

and immune structures, while reducing activation over visually salient but clinically less informative background regions.

In Fig. 4(a), compared with MM-DLS, Multi-Cog, DenseMoE, and KGT, our Grad-CAM responses are more spatially aligned with the clinician-annotated regions across HC, DED, and NCP examples. The difference is particularly evident in abnormal cases, where competing methods often produce broader or fragmented activations, whereas our model concentrates more strongly on localized nerve alterations. This suggests that the learned representation relies more on disease-related structural evidence rather than non-specific image appearance.

Fig. 4(b) further evaluates clinically relevant corneal nerve loss in systemic neuropathy. Nerves from the held-out internal test sets and complete external cohort were segmented using CCM-Pro (Li et al., 2025b), from which CNFD, CNBD, and CNFL were quantified following established clinical definitions (Alam et al., 2017). Group differences were assessed using the Kruskal–Wallis test with Bonferroni-corrected pairwise comparisons; all three metrics decreased progressively from HC to DPN− and DPN+ ($p < 0.001$). As a representative example, median CNFL decreases from 24.33 to 18.90 and 13.90 mm/mm^2, respectively. This monotonic pattern is consistent with progressive corneal nerve degeneration in diabetic peripheral neuropathy (Zhang et al., 2025; Gad et al., 2022), providing complementary evidence that the model emphasizes clinically relevant neural phenotypes.

4.3. Ablation study

We conducted progressive ablations across six clinical objectives to examine the contribution of each innovation and its key components (Fig. 5). Within each module, components are introduced sequentially on top of the preceding configuration, while the macro-level setting progressively integrates the three modules from the visual backbone to the complete framework. This design exposes the marginal contribution of each step rather than only comparing the final model with a single baseline. Overall, performance generally improves as the components are progressively introduced across diagnosis, regression, and prognosis.

Starting from the visual backbone (V0), clinically induced relation alignment, HSIC-based target-relevance filtering, and KCI-based conditional filtering with top-ρ retention are added progressively to form V1–V3. The most pronounced gain appears in ocular regression, where MAE decreases from 2.153 to 1.699. This improvement is consistent with the role of relation-guided extraction: clinically induced relations first regularize acquisition- and cohort-sensitive feature dependencies, while subsequent HSIC/KCI filtering retains factors that remain informative for the specific task. Fig. 5 further shows consistent improvements across the remaining objectives, supporting the effectiveness of task-specific robust feature extraction.

T0–T3 progressively introduce patient-semantic encoding, semantic-guided visual contextualization, and the residual contextualization pathway on top of the task-specific visual representation. A representative effect is observed in systemic regression, where MAE decreases from 1.502 to 1.188. The gain indicates that patient semantics provide complementary context for interpreting visually similar CCM patterns, thereby reducing ambiguity that cannot be resolved from image features alone. Consistent trends across the other tasks are shown in Fig. 5.

H0–H3 progressively replace a static prediction head with task conditioning, joint patient–task conditioning, and dynamic parameter generation. The clearest improvement is observed in systemic regression, where MAE decreases from 1.188 to 0.820. This result supports the central role of the hypernetwork: rather than forcing heterogeneous diagnosis, regression, and prognosis objectives to share a fixed prediction function, the model adapts its parameters to both patient context and task identity. Fig. 5 shows that this benefit is consistently preserved across the remaining objectives.

M0–M3 progressively integrate Relational Extraction, Semantic Contextualization, and Hypernetwork Prediction. Performance progressively improves as the modules are integrated; for example, ocular diagnosis AUC increases from 0.854 to 0.950 from M0 to M3. This macro-level progression indicates that the three modules address complementary sources of difficulty—visual nuisance variation, patient-level clinical heterogeneity, and task-level prediction heterogeneity—rather than providing redundant improvements. Together, the progressive trends in Fig. 5 validate the complete design from robust task-specific feature construction to patient-aware contextualization and adaptive prediction.

4.4. Computational complexity analysis

As shown in Fig. 6, the proposed framework maintains a low model complexity while achieving superior predictive performance. Under the complete-context setting (0% CD), Ours uses 43.29M parameters, substantially fewer than representative baselines such as MM-MIL (83.44M) and KGT (110.48M),

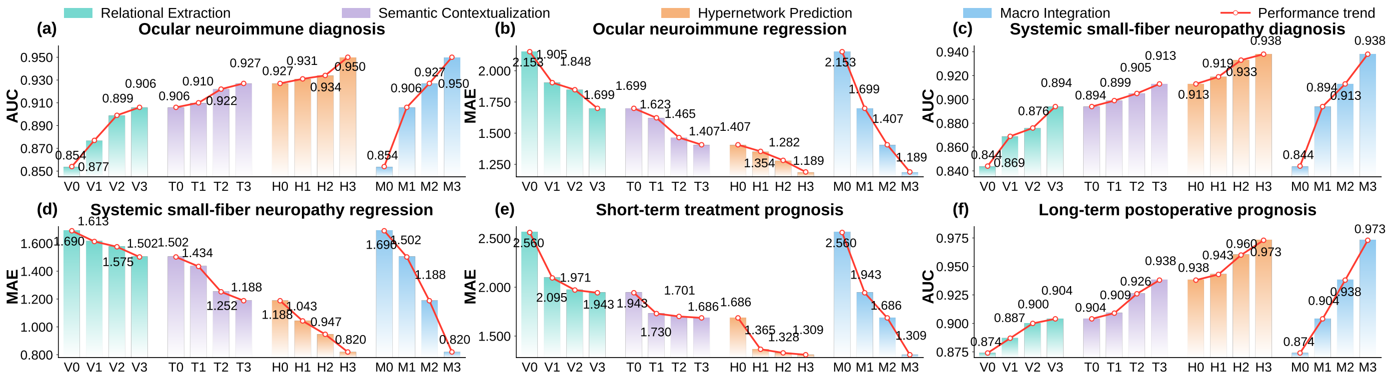

Fig. 5. Progressive ablation analysis of the proposed framework on the internal validation cohort. Component- and module-level contributions across six clinical objectives. Under the progressive construction, V3=T0, T3=H0, and H3=M3.

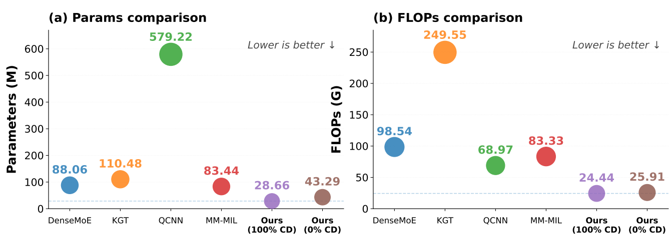

Fig. 6. Computational complexity comparison under complete-context (0% CD) and image-only (100% CD) configurations.

while requiring only 25.91G FLOPs. In the image-only configuration (100% CD), the ClinicalBERT-based clinical-context branch is removed from the active inference pathway, reducing the active model complexity to 28.66M parameters and 24.44G FLOPs. These results indicate that the performance gains are not driven by model-scale expansion: the framework relies on shared representations and lightweight patient–task-conditioned parameter generation rather than separate large task-specific networks, yielding a favorable performance–complexity trade-off and practical computational efficiency.

4.5. Hyperparameter sensitivity analysis

The best overall configuration was obtained with $\lambda_{\text{prior}} = 1.0$, $\rho = 30\%$, and $\lambda_{\text{hyper}} = 5 \times 10^{-4}$ (Fig. 7). Across all task groups, performance generally improved toward intermediate settings and declined when the corresponding constraint became excessive. Specifically, moderate λ_{prior} balances relational guidance and visual flexibility, $\rho = 30\%$ preserves informative task-specific factors while suppressing redundant channels, and moderate λ_{hyper} provides sufficient regularization without limiting patient–task-adaptive prediction. These trends indicate a stable trade-off among relation alignment, feature selection, and hypernetwork adaptation.

5. Discussion and Further Analysis

Beyond overall predictive performance, we further examined why the proposed framework is effective and whether its learned outputs remain clinically meaningful. Specifically, the discussion focuses on three complementary aspects: mechanism and robustness analyses to explain the contribution of the

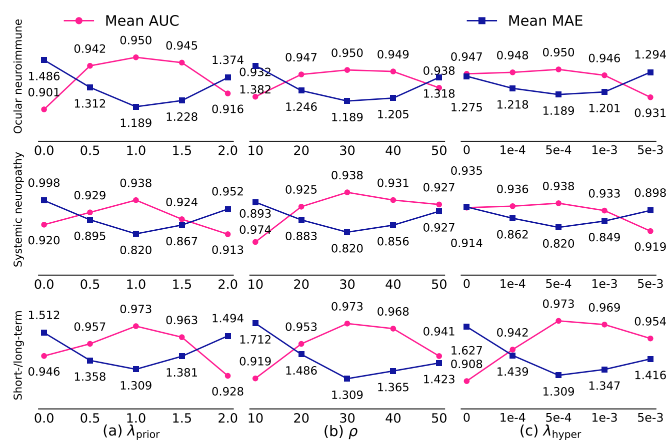

Fig. 7. Hyperparameter sensitivity across clinical tasks on the internal validation cohort. Effects of λ_{prior}, ρ, and λ_{hyper} on ocular neuroimmune, systemic neuropathy, and short-/long-term performance.

core design under task and acquisition variability, representation visualization to assess whether disease-related features become more discriminative, and clinical validity analyses to determine whether the predicted reference metrics and prognostic outputs preserve established disease-related patterns and longitudinal relevance. Together, these analyses connect model behavior with both methodological effectiveness and clinical plausibility.

5.1. Mechanism and robustness analyses

Fig. 8 examines three properties underlying the effectiveness of the proposed framework: whether clinically induced relations provide meaningful rather than generic regularization, whether patient–task adaptation reduces interference among heterogeneous objectives, and whether relation-guided feature extraction improves resistance to acquisition degradation.

Fig. 8(a) tests whether the gain from relation-guided feature extraction originates from meaningful clinical structure. The *No* setting removes relation alignment, whereas *Random* replaces the clinical relation structure with an independently sampled symmetric matrix and *Shuffled* disrupts variable correspondence while preserving its value distribution; *Relational (Ours)* uses the clinically induced relations defined in Sec. 2.1.2. The clearest effect is observed for short-term prognosis, where MAE

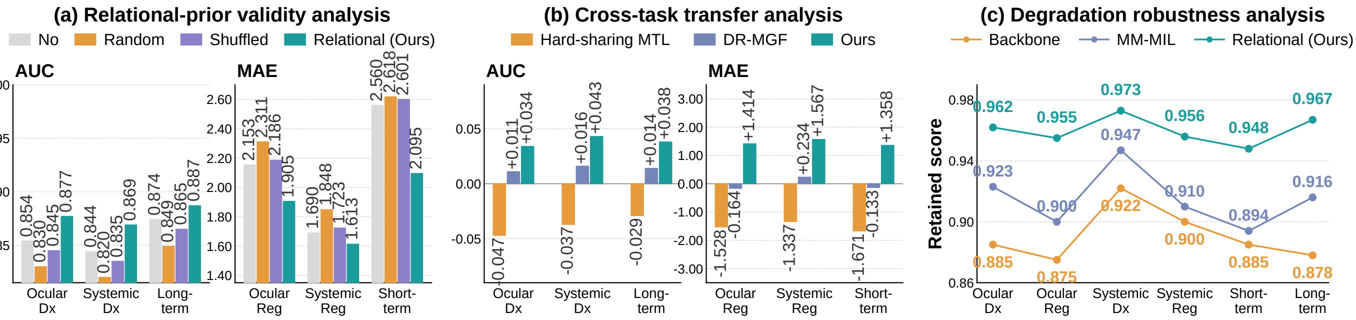

Fig. 8. Mechanism and robustness analyses. (a) Validity of the clinically induced relation structure across diagnosis, regression, and prognosis tasks. (b) Cross-task transfer relative to single-task learning. (c) Performance retention under input degradation for the backbone, MM-MIL, and relational feature extraction.

decreases from 2.560 without relation guidance to 2.095 with the clinically induced structure, while both random and shuffled relations fail to reproduce this improvement. Fig. 8(a) shows the same tendency across the remaining objectives. This contrast indicates that the benefit is not explained by imposing an arbitrary relational constraint; rather, preserving clinically supported dependencies provides a more informative organization of the latent visual space.

Fig. 8(b) evaluates whether the proposed patient–task-conditioned prediction mechanism alleviates interference among diagnosis, regression, and prognosis objectives. Cross-task transfer is defined as Δ = Method − Single for AUC-based tasks and Δ = Single − Method for MAE-based tasks, where *Method* denotes each evaluated multi-task approach and positive values indicate improvement. Hard-sharing multi-task learning produces negative transfer across all six tasks, whereas the gradient-conflict-oriented DR-MGF (Sun et al., 2026) baseline alleviates part of this interference but does not consistently yield positive transfer. In contrast, our framework improves all six objectives relative to their single-task counterparts; for example, systemic regression shows a gain of +1.567. This consistent transfer supports the proposed design in which task-specific feature selection limits inappropriate representation sharing and patient–task conditioning further adapts the prediction function to heterogeneous clinical objectives.

Fig. 8(c) further tests whether relation-guided feature extraction stabilizes representations under simulated acquisition degradation, including blur, noise, illumination variation, and contrast perturbation. Performance retention is defined as $R = M_{\text{deg}}/M_{\text{clean}}$ for AUC-based tasks and $R = M_{\text{clean}}/M_{\text{deg}}$ for MAE-based tasks, such that higher values consistently indicate better preservation. A representative example is ocular regression, for which the retained score increases from 0.875 with the backbone and 0.900 with MM-MIL to 0.955 with *Relational (Ours)*. The same ordering is observed across the six objectives in Fig. 8(c). These results are consistent with the intended role of clinically induced relation alignment: constraining feature dependencies toward stable clinical structure reduces reliance on degradation-sensitive visual correlations and improves representation stability under suboptimal imaging conditions.

Together, these analyses provide complementary mechanistic evidence for the framework: clinically induced relations organize the visual representation in a meaningful manner, patient–task adaptation mitigates heterogeneous-task interference, and relation-guided feature extraction improves robustness to acquisition degradation.

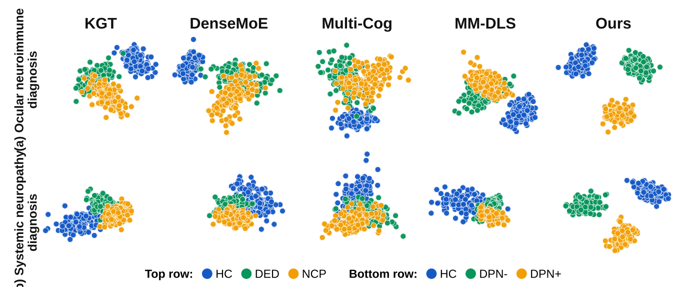

Fig. 9. t-SNE visualization of learned representations. Top: ocular neuroimmune diagnosis; bottom: systemic neuropathy diagnosis.

5.2. *t-SNE visualization*

Fig. 9 examines whether the proposed framework improves the organization of disease-related representations rather than only the final prediction scores. Across both ocular neuroimmune and systemic neuropathy diagnosis, Ours produces more compact intra-class clusters and clearer inter-class separation than KGT, DenseMoE, Multi-Cog, and MM-DLS. The difference is particularly evident for clinically adjacent phenotypes: DED and NCP remain substantially overlapped for several competing methods, whereas the proposed framework separates HC, DED, and NCP more distinctly; a similar reduction in overlap is observed among HC, DPN−, and DPN+. This representation-level separation is consistent with the intended role of task-specific feature extraction and patient-semantic contextualization, suggesting that the performance gains arise from more discriminative phenotype representations rather than solely from changes in the prediction head.

5.3. *Clinical validity of metric regression and prognosis*

Beyond numerical regression accuracy, we examined whether the predicted reference metrics reproduce clinically expected disease patterns (Fig. 10).

As shown in Fig. 10 (a), for DED severity, predicted CFS increased progressively from HC to severe DED, whereas TBUT and SIT decreased, consistent with established associations between ocular-surface signs and increasing DED severity (Sullivan et al., 2010; Begley et al., 2003).

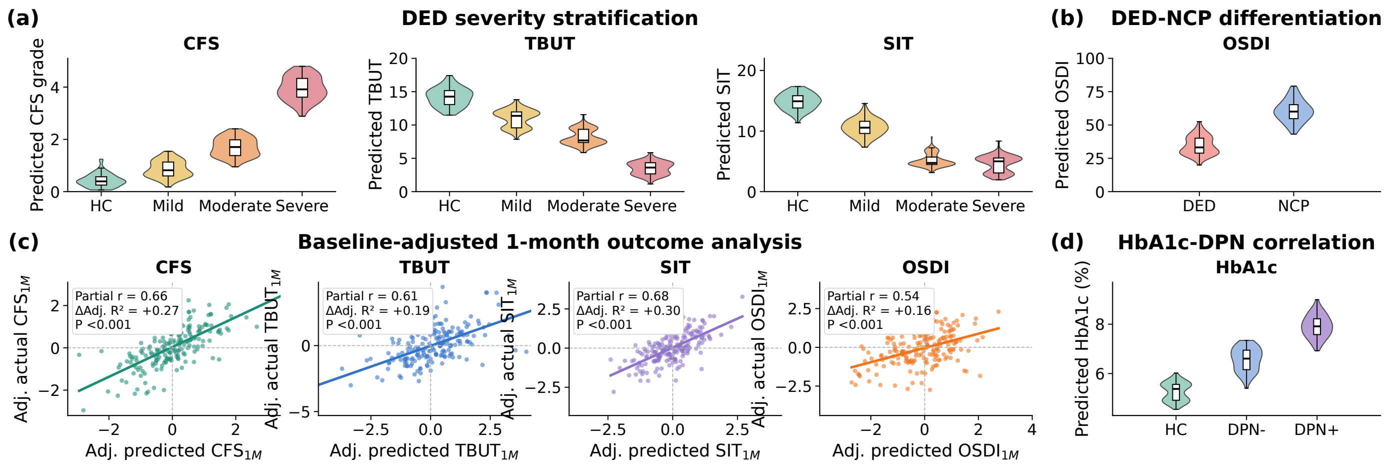

Fig. 10. Clinical validity of metric regression and prognosis. Associations with disease severity, phenotype differentiation, longitudinal outcomes, and systemic metabolic status are evaluated.

As shown in Fig. 10 (b), predicted OSDI also remained substantially higher in NCP than in conventional DED, consistent with the greater symptom burden reported in neuropathic corneal pain (Chin et al., 2024). Importantly, these severity and phenotype labels were independently established by unanimous consensus of three senior clinicians, and were not used to construct the corresponding predictions. The observed trends therefore provide an independent assessment of clinical concordance.

As shown in Fig. 10 (c), we further evaluated whether the predicted one-month outcomes retained information beyond the corresponding baseline measurements. Image-level predictions from the same patient were first averaged to obtain a single patient-level estimate. For each endpoint, the observed one-month value y_{1M} was then modeled using both the actual baseline value y_0 and the held-out model prediction $\hat{y}_{1M}$, i.e., $y_{1M} = \beta_0 + \beta_1 y_0 + \beta_2 \hat{y}_{1M} + \epsilon$. Partial correlations were computed after adjustment for y_0, and the increase in adjusted R^2 over the baseline-only model quantified the additional explanatory value of the prediction. As shown in the figure, all four endpoints, including CFS, TBUT, SIT, and OSDI, remained significantly associated with their observed one-month outcomes after baseline adjustment, indicating that the model captured longitudinal information beyond baseline status.

As shown in Fig. 10 (d), predicted HbA1c increased from HC to DPN− and DPN+, consistent with the established association between glycemic burden and diabetic peripheral neuropathy (Braffett et al., 2020; Pinto et al., 2020). Collectively, these findings suggest that the predicted reference metrics preserve clinically plausible ordering, phenotype differences, and longitudinal relevance.

6. Conclusion

We addressed the challenge of unified CCM-based assessment across ocular neuroimmune and systemic neuropathy diagnosis, continuous clinical metric regression, and longitudinal prognosis. The proposed framework combines relation-guided task-specific feature extraction, patient-semantic-guided feature contextualization, and patient–task-conditioned hypernetwork prediction to learn robust task-relevant representations while adapting prediction to individual patients and heterogeneous clinical objectives. Across internal and independent external cohorts, the framework achieved consistently strong performance across categorical, continuous, and prognostic tasks, while the predicted clinical metrics preserved clinically meaningful disease-related patterns. These findings support the potential of CCM as a unified source of imaging-derived biomarkers for disease assessment, quantitative clinical characterization, and longitudinal prediction.

A key limitation is that the clinical objectives were evaluated in task-specific cohorts rather than in a single cohort undergoing the complete set of assessments. Future work should therefore focus on prospective validation in unified, more heterogeneous patient cohorts to further establish generalizability and clinical utility.

7. Acknowledgement

This work is supported by the National Natural Science Foundation of China (62422122, 62272444, 62401551), Key research and development program of Zhejiang Province (2024C03204, 2023C04017, 2024C03101), the Project of Ningbo Public Welfare Science and Technology (2023S012, 2024S058), Ningbo 2025 S&T Mega project (2022Z134).

References

Alam, U., Anson, M., Meng, Y., Preston, F., Kirthi, V., Jackson, T.L., Nderitu, P., Cuthbertson, D.J., Malik, R.A., Zheng, Y., Petropoulos, I.N., 2022. Artificial intelligence and corneal confocal microscopy: The start of a beautiful relationship. J. Clin. Med. 11, 6199. doi:10.3390/jcm11206199.

Alam, U., Jeziorska, M., Petropoulos, I.N., Asghar, O., Fadavi, H., Ponirakis, G., Marshall, A., Tavakoli, M., Boulton, A.J.M., Efron, N., Malik, R.A., 2017. Diagnostic utility of corneal confocal microscopy and intra-epidermal nerve fibre density in diabetic neuropathy. PLoS One 12, e0180175. doi:10.1371/journal.pone.0180175.

Alsentzer, E., Murphy, J.R., Boag, W., Weng, W.H., Jin, D., Naumann, T., McDermott, M.B.A., 2019. Publicly available clinical bert embeddings, in: Proceedings of the 2nd Clinical Natural Language Processing Workshop, Association for Computational Linguistics. pp. 72–78. doi:10.18653/v1/W19-1909.

Begley, C.G., Chalmers, R.L., Abetz, L., Venkataraman, K., Mertzanis, P., Caffery, B.A., Snyder, C., Edrington, T., Nelson, D., Simpson, T., 2003. The relationship between habitual patient-reported symptoms and clinical signs among patients with dry eye of varying severity. Investig. Ophthalmol. Vis. Sci. 44, 4753–4761. doi:10.1167/iovs.03-0270.

Braffett, B.H., Gubitosi-Klug, R.A., Albers, J.W., Feldman, E.L., Martin, C.L., White, N.H., Orchard, T.J., Lopes-Virella, M., Lachin, J.M., Pop-Busui, R., 2020. Risk factors for diabetic peripheral neuropathy and cardiovascular autonomic neuropathy in the diabetes control and complications trial/epidemiology of diabetes interventions and complications study. Diabetes 69, 1000–1010. doi:10.2337/db19-1046.

Brand, L., Nichols, K., Wang, H., Shen, L., Huang, H., 2020. Joint multi-modal longitudinal regression and classification for alzheimer's disease prediction. IEEE Trans. Med. Imaging 39, 1845–1855. doi:10.1109/TMI.2019.2958943.

Chen, W., Liao, D., Deng, Y., Hu, J., 2024. Development of a transformer-based deep learning algorithm for diabetic peripheral neuropathy classification using corneal confocal microscopy images. Front. Cell Dev. Biol. 12, 1484329. doi:10.3389/fcell.2024.1484329.

Chin, J.Y., Tong, L., Liu, C., Lee, I.X.Y., Wong, J.H.F., Wong, R.K.T., Mehta, J.S., Liu, Y.C., 2024. Quality of life and symptomatology in neuropathic corneal pain in comparison with dry eye syndrome. Cornea 44, 825–831. doi:10.1097/ICO.0000000000003674.

Dieckmann, G., Goyal, S., Hamrah, P., 2017. Neuropathic corneal pain: Approaches for management. Ophthalmology 124, S34–S47. doi:10.1016/j.ophtha.2017.08.004.

Ding, R., Lu, H., Liu, M., 2025. Denseformer-moe: A dense transformer foundation model with mixture of experts for multi-task brain image analysis. IEEE Trans. Med. Imaging 44, 4037–4048.

Gad, H., Petropoulos, I.N., Khan, A., Ponirakis, G., MacDonald, R., Alam, U., et al., 2022. Corneal confocal microscopy for the diagnosis of diabetic peripheral neuropathy: A systematic review and meta-analysis. J. Diabetes Investig. 13, 134–147. doi:10.1111/jdi.13643.

Gretton, A., Bousquet, O., Smola, A.J., Schölkopf, B., 2005. Measuring statistical dependence with hilbert-schmidt norms, in: International Conference on Algorithmic Learning Theory, pp. 63–77. doi:10.1007/11564089_7.

Huang, Z., Pan, W., Alishetti, S., Beecy, A.N., Liu, Z., Gong, A., Shin, S., Clerkin, K.J., Goldsmith, R.L., Majure, D.T., Kelsey, C., vanMaanan, D., Ruhl, J., Tesfuzigta, N., Lancet, E., Kumaraiah, D., Sayer, G., Estrin, D., Weinberger, K., Uriel, N., Wang, F., 2026. Multimodal multi-instance learning for cardiopulmonary exercise testing performance prediction. npj Digit. Med. 9, 304. doi:10.1038/s41746-026-02493-w.

Koseoglu, N.D., Chen, E., Tuwani, R., Kompa, B., Cox, S.M., Cuneyt Ozmen, M., Massaro-Giordano, M., Beam, A.L., Hamrah, P., 2025. Development and validation of a deep learning model for diagnosing neuropathic corneal pain via in vivo confocal microscopy. npj Digit. Med. 8, 277.

Lao, H., Zhang, X., Tang, Y., Initiative, A.D.N., et al., 2025. Multi-tasks diagnosis of alzheimer's disease based on multi-modalities images brain networks and quaternion convolutional neural networks. Expert Syst. Appl. 281, 127607.

Li, H., Dong, A., Zhang, T., Zhou, S., Zheng, Y., Mou, L., Zhao, Y., 2025a. Morphoboost: Morphology-driven boundary enhancement model for accurate segmentation of langerhans cells in corneal confocal microscopy images, in: Int. Conf. Med. Image Comput. Comput.-Assist. Interv., Springer. pp. 382–391.

Li, H., Ma, B., Mou, L., Liu, Y., Zheng, Q., Qi, H., Zhao, Y., 2025b. Progressive distillation for incremental learning in corneal confocal microscopy segmentation. IEEE Trans. Med. Imaging .

Li, H., Mou, L., Hao, J., Zhang, J., Zhao, Y., 2025c. Dilation-assisted contrastive learning for corneal nerve segmentation in corneal confocal microscopy, in: IEEE Int. Symp. Biomed. Imaging, IEEE. pp. 1–5.

Liu, X., Dai, F., Dai, J., Wang, H., Li, Q., Wen, Y., Guo, H., Zhao, L., Wang, H., Xu, K., Weng, M., Yao, S., Lu, H., He, Y., 2026. Multimodal multitask deep learning for grading management system in non-small cell lung cancer. Nat. Commun. 17, 7579. doi:10.1038/s41467-026-74474-2.

Ma, D., Pabalan, C., Rajagopal, A., Akanksha, A., Interian, Y., Yang, Y., Raj, A., 2026. Predicting categorical and continuous alzheimer's disease outcomes from a single mri scan. Nat. Aging , 1–17.

Mineo, R., Salanitri, F.P., Bellitto, G., Kavasidis, I., De Filippo, O., Millesimo, M., De Ferrari, G.M., Aldinucci, M., Giordano, D., Palazzo, S., et al., 2024. A convolutional-transformer model for ffr and ifr assessment from coronary angiography. IEEE Trans. Med. Imaging 43, 2866–2877.

Moein, H.R., Akhlaq, A., Dieckmann, G., Abbouda, A., Pondelis, N., Salem, Z., Müller, R.T., Cruzat, A., Cavalcanti, B.M., Jamali, A., Hamrah, P., 2020. Visualization of microneuromas by using in vivo confocal microscopy: An objective biomarker for the diagnosis of neuropathic corneal pain? Ocul. Surf. 18, 651–656. doi:10.1016/j.jtos.2020.07.004.

Mou, L., Qi, H., Liu, Y., Zheng, Y., Matthew, P., Su, P., Liu, J., Zhang, J., Zhao, Y., 2022. DeepGrading: deep learning grading of corneal nerve tortuosity. IEEE Trans. Med. Imaging 41, 2079–2091.

Pinto, M.V., Rosa, L.C.G.F., Pinto, L.F., Dantas, J.R., Salles, G.F., Zajdenverg, L., Rodacki, M., Lima, M.A., 2020. Hba1c variability and long-term glycemic control are linked to peripheral neuropathy in patients with type 1 diabetes. Diabetol. Metab. Syndr. 12, 85. doi:10.1186/s13098-020-00594-4.

Preston, F.G., Meng, Y., Burgess, J., Ferdousi, M., Azmi, S., Petropoulos, I.N., et al., 2022. Artificial intelligence utilising corneal confocal microscopy for the diagnosis of peripheral neuropathy in diabetes mellitus and prediabetes. Diabetologia 65, 457–466. doi:10.1007/s00125-021-05617-x.

Sullivan, B.D., Whitmer, D., Nichols, K.K., Tomlinson, A., Foulks, G.N., Geerling, G., Pepose, J.S., Kosheleff, V., Porreco, A., Lemp, M.A., 2010. An objective approach to dry eye disease severity. Investig. Ophthalmol. Vis. Sci. 51, 6125–6130. doi:10.1167/iovs.10-5390.

Sun, Y., Hu, X., Xu, X., Li, J., Shi, Y., Zeng, L.L., 2026. Learning task-preferred inference routes for gradient de-conflict in multi-output dnns. IEEE Trans. Pattern Anal. Mach. Intell. 48, 3154–3166. doi:10.1109/TPAMI.2025.3635844.

Wolffsohn, J.S., Arita, R., Chalmers, R., Djalilian, A., Dogru, M., Dumbleton, K., Gupta, P.K., Karpecki, P., Lazreg, S., Pult, H., Sullivan, B.D., Tomlinson, A., Tong, L., Villani, E., Yoon, K.C., Jones, L., Craig, J.P., 2017. TFOS DEWS II diagnostic methodology report. Ocul. Surf. 15, 539–574. doi:10.1016/j.jtos.2017.05.001.

Zhang, K., Peters, J., Janzing, D., Schölkopf, B., 2011. Kernel-based conditional independence test and application in causal discovery, in: Proceedings of the 27th Conference on Uncertainty in Artificial Intelligence, AUAI Press. pp. 804–813.

Zhang, X., Han, L., Shi, Y., Zheng, Y., Alam, U., Ferdousi, M., Malik, R., 2025. Hmsvit: A hierarchical masked self-supervised vision transformer for corneal nerve segmentation and diabetic neuropathy diagnosis. arXiv preprint arXiv:2506.19474 .

Zhao, J., Ding, R., Liu, M., 2026. Kgt: Knowledge-guided graph transformer for neurodegenerative disease diagnosis and brain age prediction with mri. Med. Image Anal. 113, 104143.

Zhao, Y., Zhang, J., Pereira, E., Zheng, Y., Su, P., Xie, J., Zhao, Y., Shi, Y., Qi, H., Liu, J., et al., 2020. Automated tortuosity analysis of nerve fibers in corneal confocal microscopy. IEEE Trans. Med. Imaging 39, 2725–2737.